

Biologic Considerations

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Success in clinical dentistry requires a thorough understanding of the anatomical and biologic nature of the tooth, with its components of enamel, dentin, pulp, and cementum, as well as the supporting tissues of bone and gingiva (Fig 1-1; see also Fig 1-9a). Dentistry that violates the physical, chemical, and biologic parameters of tooth tissues can lead to premature restoration failure, compromised coronal integrity, recurrent caries, patient discomfort, or even pulpal necrosis.

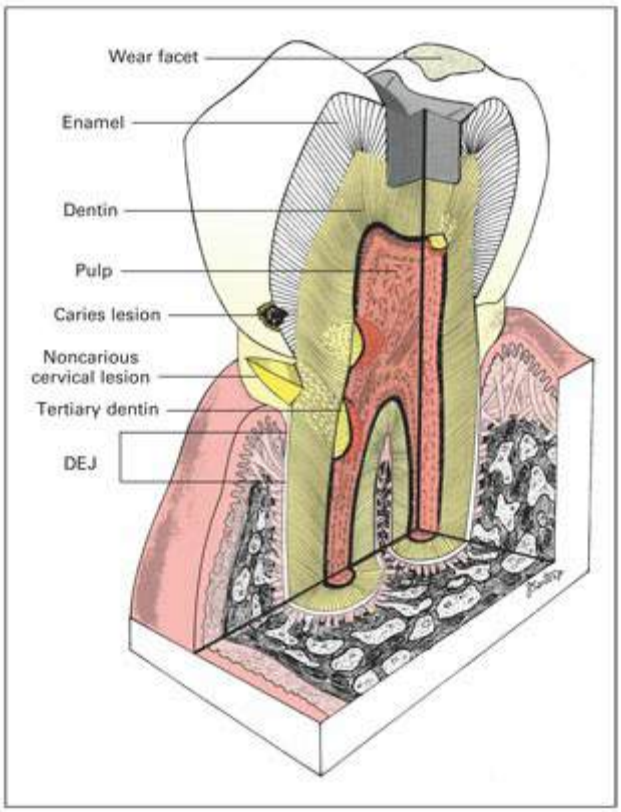


Fig 1-1 Component tissues and supporting structures of the tooth. DEJ—dentinoenamel junction.

The principles, materials, and techniques that constitute operative dentistry are effective only when utilized within a framework based on these biologic parameters. This chapter presents a morphologic and histologic review of tooth tissues with emphasis on their clinical significance for the practice of restorative dentistry.

Enamel

Enamel provides the shape and hard, durable outer surface of teeth, which protects the underlying dentin and pulp (see Fig 1-9a). Both color and form contribute to the esthetic appearance of enamel. Much of the art of restorative dentistry comes from efforts to simulate the color, texture, translucency, and contours of enamel with synthetic dental materials, such as resin composite or porcelain. Nevertheless, the lifelong preservation of the patient's own enamel is one of the defining goals of the discipline of operative dentistry. Although enamel is capable of lifelong service, its crystalline mineral makeup and rigidity, exposed to an oral environment of occlusal, chemical, and bacterial challenges, make it vulnerable to acid demineralization, attrition (wear), and fracture (Fig 1-2). Mature enamel is unique compared with other tissues because, besides alterations in its mineral content, repair or

replacement can only be accomplished through dental therapy.

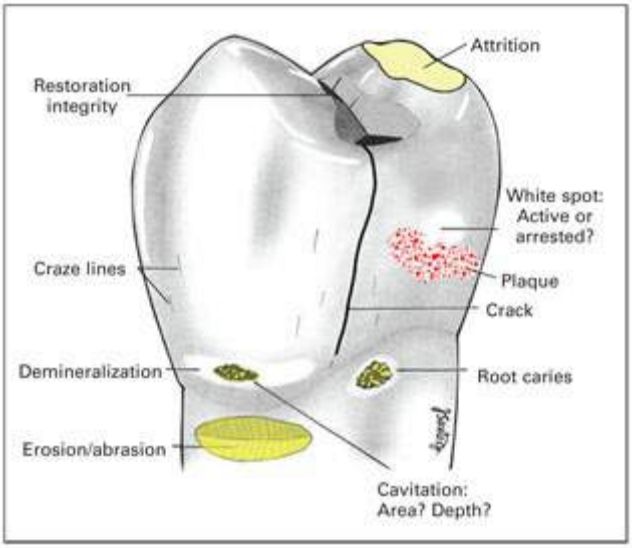


Fig 1-2 Observations of clinical importance on the tooth surface.

Permeability

At maturity, enamel is 96% inorganic hydroxyapatite mineral by weight and more than 86% hydroxyapatite mineral by volume. Enamel also contains a small volume of organic matrix, as well as 4% to 12% by volume water, which is contained in the intercrystalline spaces and in a network of micropores opening to the external surface.[1] These microchannels form a dynamic connection between the oral cavity and the pulpal interstitial space and dentinal tubule fluids.[2] Various fluids, ions, and low–molecular weight substances, whether deleterious, physiologic, or therapeutic, can diffuse through the semipermeable enamel. Therefore, the dynamics of acid demineralization, reprecipitation or remineralization, fluoride uptake, and vital bleaching therapy are not limited to the surface but are active in three dimensions.[3–6] When teeth become dehydrated, as from nocturnal mouth breathing or rubber dam isolation for dental treatment, the empty micropores make the enamel appear chalky and lighter in color (Fig 1-3). The condition is reversible with return to the "wet" oral environment. There is some evidence that the permeability of the enamel decreases with age and may be affected by various dental procedures, such as tooth whitening, acid etching, or the physical removal of the outermost layer of enamel.[7–9]



Fig 1-3 Color change resulting from dehydration. The right central incisor was isolated by rubber dam for approximately 5 minutes. Shade matching of restorative materials should be determined with full-spectrum lighting before isolation.

Lifelong exposure of semipermeable enamel to the ingress of elements from the oral environment into the mineral structure of the tooth results in coloration intensity and resistance to demineralization. The yellowing of older teeth may be attributed to thinning or increased translucency of enamel, accumulation of trace elements in the enamel structure, and perhaps the sclerosis of mature dentin. This yellowing may be treated conservatively with at-home or in-office bleaching. The enamel remineralization process benefits from the incorporation of fluoride from water sources or toothpaste and from the fluoride concentrated in the biofilm (plaque) that adheres to enamel surfaces. Enamel damaged by acid-producing biofilm bacteria can be repaired by remineralization with fluoride, which increases the rate of conversion of hydroxyapatite into more stable and less acid-soluble crystals of fluorohydroxyapatite or fluoroapatite.[10] There has been a considerable amount of research recently directed at further enhancing the effectiveness of fluoride remineralization by creating new delivery systems that increase the available calcium and phosphate required to form fluoro-hydroxyapatite and fluoroapatite.[11] With aging, color (hue) is intensified, but acid solubility of enamel, pore volume, water content, and permeability are reduced, although a basic level of permeability is maintained.[12]

Clinical appearance and defects

The dentist must pay close attention to the surface characteristics of enamel for evidence of pathologic or traumatic conditions. Key diagnostic signs include color changes associated with demineralization, cavitation, excessive wear, morphologic faults or fissures, and cracks (see Fig 1-2).

Color

Enamel translucency is directly related to the degree of mineralization, and its color is primarily a function of its thickness and the color of the underlying dentin. From approximately 2.5 mm at cusp tips and 2.0 mm at incisal edges, enamel thickness decreases significantly below deep occlusal fissures and tapers to become very thin in the cervical area near the cementoenamel junction (CEJ). Therefore, the young anterior tooth has a translucent gray or slightly bluish hue near the incisal edge. A more chromatic yellow-orange shade predominates cervically, where dentin shows through thinner enamel. Coincidentally, in about 10% of teeth, a gap between enamel and cementum in the cervical area leaves vital, potentially sensitive dentin completely exposed.[13]

Anomalies of development and mineralization, extrinsic stains, antibiotic therapy, and excessive fluoride can alter the natural color of the teeth.[14] However, because caries is the primary disease threat to the dentition, enamel discoloration related to demineralization caused by acid from a few microorganisms, primarily mutans streptococci, within biofilm[15] is a critical diagnostic observation. Subsurface enamel porosity from demineralization is manifested clinically as a milky white opacity termed a *white spot lesion* (Figs 1-2 and 1-4). Early enamel fissure–caries lesions are difficult to detect on bitewing radiographs. However, diagnostic accuracy can be improved by a systematic visual ranking of the enamel discoloration adjacent to pits and fissures, which in turn is correlated with the histologic depth of demineralization.[16,17] In the later stages of enamel demineralization extending to near the dentinoenamel junction (DEJ), the white-spot opacity is evident not only when the tooth is air dried but also when it is wet with saliva.[18] It may take 4 to 5 years for demineralization to progress through the enamel,[19] but with improved plaque removal and remineralization, the lesion may arrest and, with time, appear normal again. In one study, 182 white spot lesions in 8-year-old children were reevaluated at age 15 years: 9% had cavitated, 26% appeared unchanged, and 51% appeared clinically sound.[20] In addition, sealing an initial caries lesion with resin has also been shown to be an effective method for arresting its further development.[21,22]

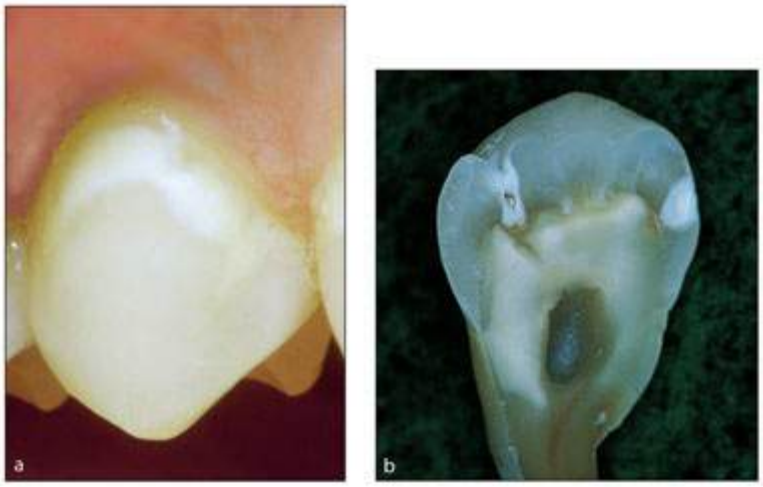


Fig 1-4 *(a)* White spot lesion on the facial surface of the maxillary premolar. *(b)* Premolar with both an occlusal fissure-caries lesion (Class 1), extending into the dentin, and a proximal smooth-surface caries lesion (Class 2).

A longstanding chalky and roughened white-spot appearance of the facial or lingual enamel surface (see Fig 1-4a) may be a result of factors such as inadequate oral hygiene, a cariogenic diet, and an insufficient amount of saliva resulting from medical conditions or medication. All of these factors place the patient at a higher risk for caries.[23] As the caries progresses, the overlying enamel takes on a blue or gray tint that provides a clinical sign indicating advanced dentin involvement. With the advent of effective remineralization, dentin bonding techniques, and fissure sealants, several authorities have suggested that invasive restorative procedures or replacement restorations should be considered only if caries lesion extension to dentin can be confirmed by visual signs of deep discoloration, enamel cavitation to dentin, or radiographic evidence.[24,25]

Cavitation

In the early stages of an enamel caries lesion, acid from the biofilm penetrates through the eroded crystal spaces to form a subsurface lesion of demineralized and porous mineral structure that appears clinically as a white spot. The acid protons follow the direction of the widened intercrystalline spaces of the affected enamel rods toward the DEJ. If the cariogenic biofilm, the etiology of the lesion, is not regularly removed through preventive measures, the lesion will progress in depth to the DEJ and into the dentin. When seen in two dimensions, as in a radiograph, smooth-surface enamel lesions are triangular, with the base of the triangle at the enamel surface; in a three-dimensional view, the proximal enamel lesion is a cone with its base equivalent in location and area to the demineralized enamel surface and

its apex closest to the DEJ. The deepest demineralized enamel rods, those at the apex of the cone, are first to be demineralized to the depth of the DEJ because of their longer time of exposure to the acid concentrations produced by the biofilm. The nature of enamel caries lesions in occlusal fissures is similar, but the shape is more complex because it occurs simultaneously at the confluence of two or more cuspal lobes, each with divergent rod directions (see Fig 1-4b). In two dimensions, a fissure-caries lesion presents with the apex of the triangular-shaped lesion located where the initial demineralization occurs simultaneously in both of the opposing internal surfaces of the occlusal fissure, and as the caries process follows the divergent rods of both opposing lobes toward the dentin, the lesion widens to form a broader base that parallels the DEJ.

Along with regular plaque removal, topical fluoride applications help to limit or even reverse enamel demineralization.[26] Some preventive materials attempt to replace minerals in the subsurface enamel lesion using home applications of amorphous and reactive calcium phosphate complexes.[27] Another product employing synthetic hydroxyapatite in an acid paste is said to repair defects and replace crystals within a matter of minutes.[28]

Unless prevention or remineralization can abort or reverse the carious demineralization, the dentin structure is compromised and can no longer support the enamel, which eventually breaks away to create a "cavity" (Fig 1-5). A restoration must then be placed. Untreated, the cavitation expands to compromise the structural strength of the crown, and microorganisms proliferate and infiltrate deep into dentin to jeopardize the vitality of the pulp. When the caries lesion extends past the CEJ, as in root caries (see Fig 1-2), factors such as isolation, access, and gingival tissue response complicate the restorative procedure.



Fig 1-5 Maxillary molar with extensive carious dentin. This is only the initial entry through unsupported enamel into the carious dentin; the final preparation of the tooth will likely remove at least the distolingual cusp and the marginal ridge to eliminate any unsupported enamel.

Wear

Enamel is as hard as steel,[29] with a Knoop Hardness Number of 343 (compared with 68 for dentin). However, enamel will wear because of attrition or frictional contact against opposing enamel or harder restorative materials, such as porcelain. The normal physiologic contact wear rate for enamel is 15 to 29 μm per year.[30] Restorative materials that replace or function against enamel should have compatible wear, smoothness, and strength characteristics. Heavy occlusal wear is demonstrated when rounded occlusal cuspal contours are ground to flat facets (see Figs 1-1 and 1-2). Depending on factors such as bruxism, other parafunctional habits, malocclusion, age, and diet, cusps may be lost completely and enamel abraded away so that dentin is exposed and occlusal function compromised (Fig 1-6). In preparing a tooth for restoration, a cavity outline form should be designed so that the margins of restorative materials avoid critical, high-stress areas of occlusal contact.[31] The potential effects of lost vertical dimension from tooth wear may be offset by active tooth eruption and apical cemento-genesis.[32,33]



Fig 1-6 Excessive occlusal enamel and dentin loss from a combination of bruxism, attrition, and erosion. (Courtesy of Van B. Haywood, Augusta, Georgia.)

Faults and fissures

Various defects of the enamel surface may contribute to the accumulation and retention of plaque. Perikymata (parallel ridges formed by cyclic deposition of enamel), pitting defects formed by termination of enamel rods, and other hypoplastic flaws are common, especially in the cervical area.[1] Limited linear defects or craze lines result from a combination of occlusal loading and age-related loss of resiliency but are generally not clinically significant. Organic films of surface pellicle and dental cuticles, extending 1 to 3 μm into the enamel, may play key roles in ion exchange and in adhesion and colonization of bacterial plaque on the enamel surface.[33,34]

Of greater concern are the fissure systems on the occlusal surfaces and, to a lesser extent, on buccal and lingual surfaces of posterior teeth. A deep fissure is formed by

incomplete fusion of lobes of cuspal enamel in the developing tooth. The resulting narrow clefts provide a protected niche for acidogenic bacteria and the nutrients they require (Fig 1-7; see also Fig 1-4b). It is estimated that caries lesions are five times more likely to occur in occlusal fissures and two and a half times more likely to occur in buccal and lingual fissures than in proximal smooth surfaces.[35] The 2000 US Surgeon General's report,[36] which was based on a national survey of dental health, confirms that overall caries experience, especially that of smooth-surface lesions, is declining. A report from the National Center for Health Statistics, based on the US National Health and Nutrition Examination Survey comparing various survey time periods from 1988–1994 through 2007–2008, indicated that the incidence of untreated caries in the overall US population has been steadily decreasing.[37] This survey found that the only segment of the population that had shown a significant increase in caries over this time period was the segment consisting of children who were 2 to 4 years of age.[38] The fissured surfaces of the teeth are relatively inaccessible for plaque-control measures and account for nearly 90% of total decayed, missing, and filled surfaces (DMFS) in US schoolchildren. Several studies offer evidence that the physical barrier provided by an enamel-bonded resin fissure sealant is an effective preventive treatment for high-caries-risk patients and for individual teeth with incipient enamel pit and fissure lesions.[39–41]

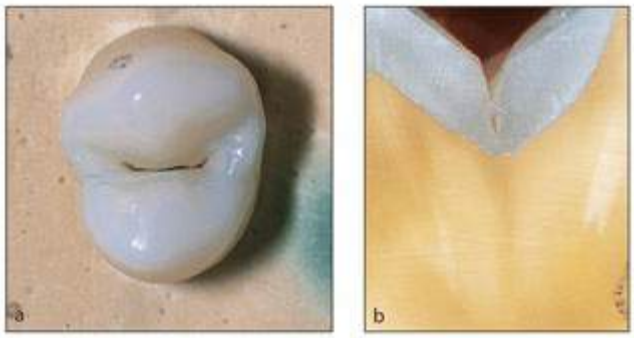


Fig 1-7 *(a)* Fissured occlusal surface of a maxillary premolar. *(b)* Cross section of the fissure shown in *a*.

Cracks

Although craze lines in the surface enamel are of little consequence, pronounced cracks that extend from developmental grooves across marginal ridges to axial surfaces, or from the margins of large restorations, may portend coronal or cuspal fracture. A crack defect is especially critical when the crack, viewed within a cavity preparation, extends through dentin or when the patient has pain while chewing (Fig 1-8). A cracked tooth that is symptomatic or involves dentin requires a restoration

that provides complete coronal coverage or at least adhesive splinting.[42,43] It should be noted, however, that even if a crack is identified early in patients with a diagnosis of reversible pulpitis and a crown is placed, subsequent root canal treatment may still be necessary in about 20% of the cases.[44]



Fig 1-8 *(a)* Molar with pronounced cracks extending across the mesial and distal marginal ridges. *(b)* Same molar with the occlusal restoration removed, exposing a mesiodistal incomplete fracture across the pulpal floor. (Courtesy of Van B. Haywood, Augusta, Georgia.)

Rod and interrod crystal structure

Enamel is a mineralized epidermal tissue. Ameloblast cells of the developing tooth secrete the organic matrix gel to define the enamel contours and initiate its mineralization. Calcium ions are transported both extra- and intracellularly to form "seeds" of hydroxyapatite throughout the developing matrix. These hydroxyapatite seeds form nidi for crystallization, and the crystals enlarge and supplant the organic matrix. The repeating molecular units of hydroxyapatite, $Ca_{10}(PO_4)_6(OH)_2$, make up the building blocks of the enamel crystal. However, the majority of apatite units exist in an impure form in which carbonate is substituted in the lattice, resulting in a destabilizing effect on the crystal. When exposed to plaque acids, the carbonated components of the crystal are the most susceptible to demineralization and the first to be solubilized. Both the therapeutic substitution of fluoride into the enamel apatite crystal and the facilitatory role of fluoride to enhance remineralization following cycles of acid dissolution are key to the dynamics of remineralization. In the presence of fluorides, enamel crystals in the incipient caries lesion are replaced or repaired with fluoroapatite or fluorohydroxyapatite, which are relatively insoluble. Therefore, the best outcome of repeated cycles of demineralization-remineralization, when accompanied by plaque control and fluoride availability, is a more caries-resistant enamel.[6]

The maturing ameloblast cell develops a cytoplasmic extension, the *Tomes' process*, which simultaneously secretes enamel protein matrix and initiates the

mineralization and orientation of enamel crystals. The divergent directions of the crystals generated from the central and peripheral surfaces of Tomes' processes, repeated in a symmetric pattern, form the two basic structural units of enamel: cylindric enamel rods and the surrounding interrod enamel. Figure 1-9 shows electron microscope photomicrographs of enamel, progressing from a macrostructural image to ultrastructural images showing individual enamel crystals.

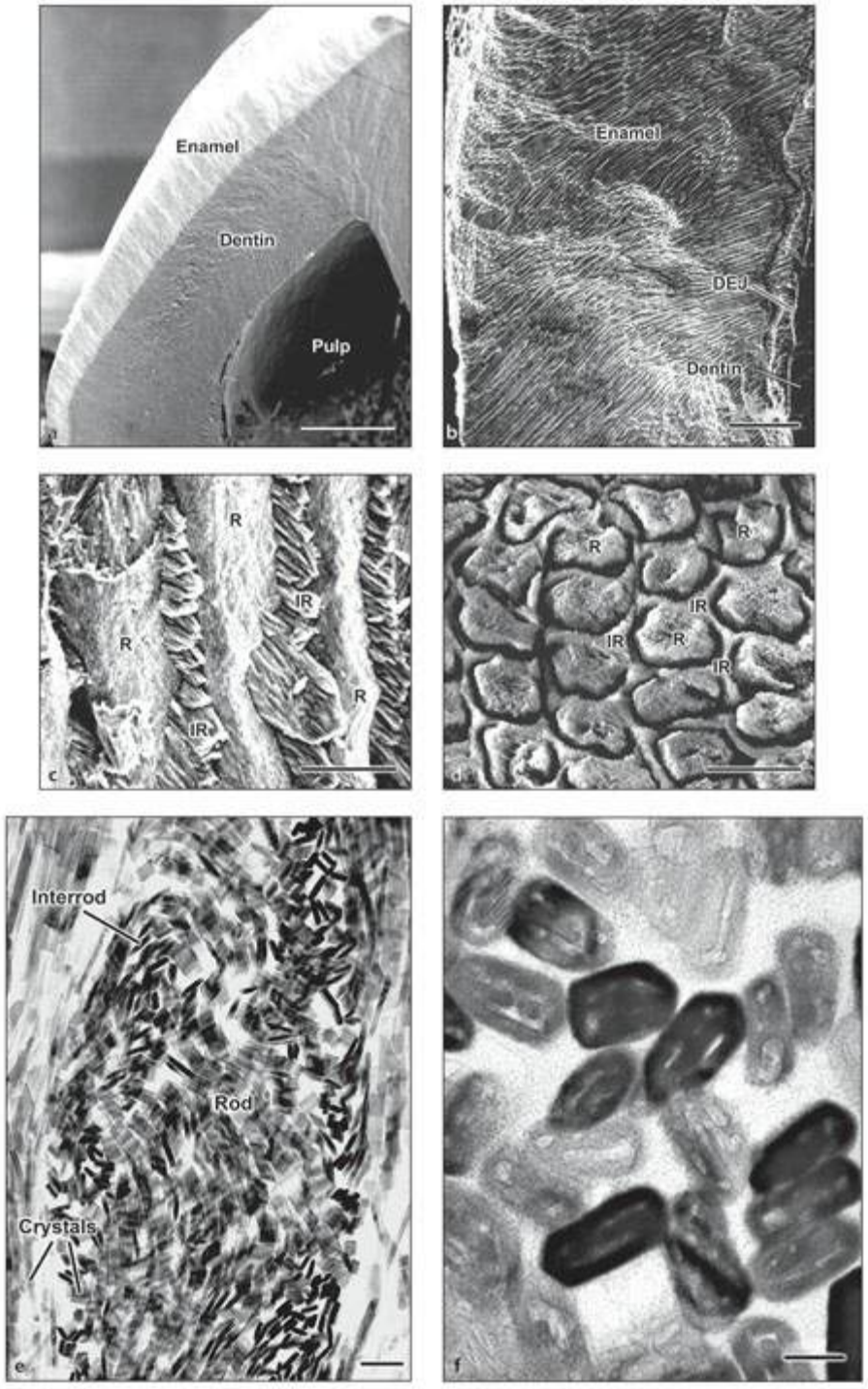


Fig 1-9 Enamel composition. *(a)* Scanning electron photomicrograph of a cross section of a tooth crown showing enamel as the outer protective covering for the tooth. (Bar = 1 mm.) *(b)* Scanning

electron photomicrograph showing the complex of enamel rods and the DEJ. (Bar = 100 µm.) *(c)* Scanning electron photomicrograph showing enamel rods (R) and interrod enamel (IR). (Bar = 6 µm.) *(d)* Scanning electron photomicrograph of a cross section of enamel rods (R) and interrod enamel (IR). Note the connecting isthmus between the two enamel components and the gap (sheath) around the rods. (Bar = 10 µm.) *(e)* Transmission electron photomicrograph showing divergent crystal orientation in rodent enamel rod and interrod enamel. (Bar = 0.1 µm.) *(f)* Transmission electron photomicrograph showing the elongated hexagonal shape of hydroxyapatite crystals in enamel. The dimensions of each crystal are in the range of 30 × 60 nm. (Bar = 20 nm.) (Reprinted from Nanci[12] with permission.)

The crystals in the enamel rods and interrod enamel differ only in the orientation of the crystals: Interrod crystals are almost perpendicular to rod crystals. In mature enamel, the closely packed, hexagonal crystals have cross-sectional dimensions of approximately 30 × 60 nm (see Fig 1-9f). The matrix proteins, enamelins, and water of hydration form a shell, or envelope, around each crystal. With the exception of the amorphous inner and outer enamel surface, the rod and interrod enamel are thought to be continuous throughout the thickness of the enamel. The multitude of crystals that form these two entities may also span the width of the enamel structure. The appearance of light and dark bands observed in sectioned specimens of enamel are known as *Hunter-Schreger bands* (Fig 1-10). This optical effect, seen under magnification in cut or fractured sections of tooth structure, is a result of the variation of light reflection from the bands of the enamel crystals that are oriented in different directions. The variation in both density and orientation of these crystals may have a direct effect on both the degree of mineral dissolution when exposed to acidic solutions as well as the susceptibility of different areas of the tooth to the development of crack lines in enamel.[45,46] The crystals within the cylinders of rod enamel run parallel to the long axis of the rods, which are approximately perpendicular to the enamel surface. A narrow space filled with organic material around three-fourths of each rod, called the *rod sheath*, separates the two enamel units. However, the two separate enamel components are connected at the portion of the rod circumference that is not bounded by the rod sheath to form an isthmus of confluent crystals (see Fig 1-9d). In cross section, the rod core and the connecting isthmus of interrod enamel together have traditionally been described as keyhole-shaped and as the basic repeating structural unit of enamel. However, recent studies show the interrod enamel to be continuous within the enamel mass and to be a step ahead of the rod in development. Therefore, the current interpretation of the structure of enamel is that of cylindric enamel rods embedded in the surrounding interrod enamel.[12]

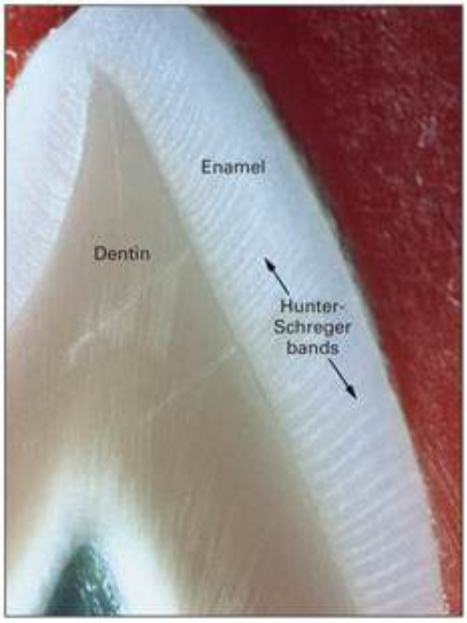


Fig 1-10 The appearance of Hunter-Schreger bands (alternate dark and light bands) viewed on the labial surface of a maxillary canine using reflected light. (Reprinted from Lynch et al[45] with permission.)

Enamel and acid etching

The spacing and divergent orientation of the crystals in the rod and in the interrod enamel make the enamel rod differentially soluble when exposed for a brief time to weak acids. Depending on the acid, contact time, and plane of cavity preparation, either the ends or the sides of the crystals may be preferentially exposed. Different etch patterns have been described depending on the type and contact time of the etchant and whether the primary dissolution affects the rod or the interrod structure.[47,48]

The initial effect of acid contact in etching enamel for bonding to restorative materials is to remove about 10 μm of surface enamel, which typically contains no rod structure. Then, with rod and interrod structure exposed, the differential dissolution of enamel rod and interrod structure forms a three-dimensional macroporosity (Fig 1-11). The acid-treated enamel surface has a high surface energy so that resin monomer flows into, intimately adapts to, and polymerizes within the pores to form retentive resin tags that are up to 20 μm deep. At the same time, the internal cores of all the exposed individual crystals are solubilized to create a multitude of microporosities. It is these countless numbers of minitags, formed within the individual crystal cores, that contribute most to the enamel-resin bond.[49] Because there are 30,000 to 40,000 enamel rods per square millimeter of a surface of cut enamel, and the etch penetration increases the bondable surface area 10- to

20-fold, the attachment of resin adhesives to enamel through micromechanical interlocking is extremely strong.[50,51]

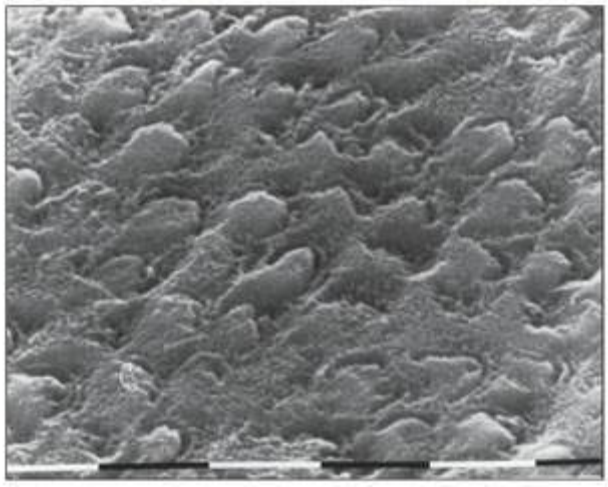


Fig 1-11 Scanning electron photomicrograph of an acid-etched enamel surface. Note the keyhole-shaped rods and uneven surface formed by the disparity in depth of rod heads and rod peripheries. (Bar = 10 μm.)

As stated, the crystals within the enamel rod cylinders run parallel to the length of the enamel rods, which are approximately perpendicular to the external enamel surface. A cavity wall preparation that is perpendicular to the surface will expose predominantly the sides of both the enamel rods and their crystals. This configuration is recommended for amalgam preparations because it preserves the dentinal support of the enamel, but it does not present the optimum bondable enamel substrate. When the transverse section or face of the crystal, rather than its side, is exposed to acid, the central core of the crystal is most susceptible to acid dissolution. Resin bond strengths are twice as high when adhering to the acid-etched ends of the crystals as compared with the sides of the crystals.[52] Thus, a tangential cut or bevel of approximately 45 degrees across a 90-degree cavosurface angle of a prepared cavity will expose the ends of the rods and their rod crystals. Beveling enamel cavosurface angles of cavity preparations for resin composite is generally recommended to expose the ends of the rods and to maximize the integrity of the restoration at its margins.[53,54] An exception is on occlusal surfaces, where beveling would extend tapering resin margins into areas of increased stress. Regardless of the variation in the etch pattern, the orientation of the enamel crystals, or the selected tooth surface, the acid-etch modification of enamel for micromechanical retention provides a conservative, reliable alternative to macromechanical undercuts traditionally used for retention of restorations.[55]

Strength and resilience

Enamel is hard and durable, but the rod sheaths, where the crystals of the interrod enamel abut three-fourths of each enamel rod cylinder, form natural cleavage lines through which longitudinal fracture may occur. The tensile bond strength of enamel rods is as low as 1¼ MPa.[56] The fracture resistance between enamel rods is weakened if the underlying dentinal support is pathologically destroyed or mechanically removed (Fig 1-12). Fracture dislodgment of the enamel rods that form the cavity wall or cavosurface margin of a dental restoration creates a gap defect. Leakage or ingress of bacteria and their by-products may lead to secondary caries lesions.[57] Some clinical dental treatments and procedures, such as whitening treatments or acid etching prior to restorative procedures, can directly affect the mechanical properties of enamel, including its hardness and modulus of elasticity.[58,59] When resin composite is adhesively bonded to approximately parallel opposing walls of a cavity preparation, stress development due to polymerization shrinkage has led to reports of enamel microcracks and crazing at margins.[60,61] Therefore, beveling acute or right-angle enamel cavosurface margins so that the bond near margins is primarily to cross-sectional rods and not to the sides of rods is believed to be beneficial in preventing these fractures.[62] Considering the variation in direction of enamel rods and interrod enamel and the structural damage caused by high-speed eccentric bur rotation, planing the cavosurface margin with hand instruments or low-speed rotary instruments to remove any friable or fragile enamel structure is recommended as a finishing step.

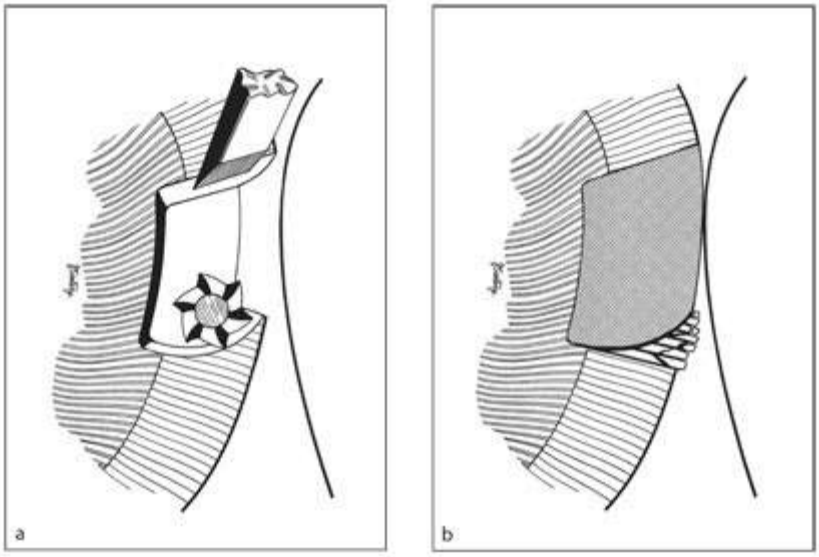


Fig 1-12 *(a)* Coronal section through an interproximal box in a cavity preparation. Use of a rotary instrument (bur), which may leave the proximal wall with an acute enamel angle and undermined enamel, requires careful planning. *(b)* Marginal defect, resulting from improper cavity wall

preparation, leads to eventual loss of enamel at the restoration interface.

Although enamel is incapable of self-repair, its protective and functional adaptation is noteworthy. Carious demineralization to the point of cavitation generally takes several years. In comparison with the underlying dentin, enamel demineralization is much slower because the apatite crystals in enamel are 10 times larger than those in dentin[63] and offer less surface-to-volume exposure to acids. The crystals are pressed so tightly together that their hexagonal shape is distorted,[12] but this tight adaptation makes for little or no space for acid penetration between the crystals. With preventive measures and exogenous or salivary renewal of calcium, phosphates, and especially fluorides, the dynamics of demineralization can be stopped or therapeutically reversed. Additionally, the crystals are separated by a thin organic matrix that provides some additional strain relief to help prevent fracture.[64] Studies on the mechanical properties of enamel indicate that the structural and compositional characteristics of the minor protein component found surrounding the enamel rods and individual hydroxyapatite crystals may significantly affect the mechanical properties of enamel.[65]

Enamel thickness and its degree of mineralization are greatest in occlusal and incisal areas of enamel where masticatory contact occurs.[66] The enamel rods are grouped in bundles that undulate in an offset pattern as they course to the surface. As a functional adaptation to occlusal stress, the spiraling weave of rod direction is so pronounced at the cusp tips of posterior teeth that it is referred to as “gnarled” enamel. If enamel were uniformly crystalline, it would shatter with occlusal function. An enamel structure with divergent crystal orientations organized into two interwoven substructures—enamel rods and interrod enamel—yet bound at a connecting area by continuous crystals provides a strong latticework. The enamel rods, which are parallel to each other and perpendicular to the surface structurally, limit the lateral propagation of occlusal stress and transfer it unidirectionally to the resilient dentinal foundation.[67]

Dentin

Dentin provides both color and an elastic foundation for the enamel. The radicular (root) dentin covered with cementum and the coronal (crown) dentin supporting the enamel form the bulk of the structure of the tooth. The strength and durability of the coronal structures are related to dentinal integrity. To the extent that open dentinal

tubules can become closed and impermeable, dentin is a protective barrier and chamber for the vital pulp tissues. As a tissue without substantive vascular supply or innervation, it is nevertheless able to respond to external thermal, chemical, or mechanical stimuli.

Dentinoenamel junction

The transition between the highly mineralized enamel and the collagen-containing dentin is a complex junction of two structurally different tissues. This interface, the DEJ, must resist fracture and separation under the extreme forces from occlusal loading. The DEJ has been described as a transitional area rather than a definite line demarcating the junction of the two tissues. Although various methods of measurement have resulted in a wide range of values for the mean width of the DEJ, a majority of studies seem to report values for the DEJ width that fall within a range of 2 to 15 µm.[68–71] It has been noted that the width of the DEJ may vary for different locations in the tooth.[68] The transitional band of the DEJ appears to be scalloped with wavelike crests pointing outward toward the enamel; there is conjecture that this interlocking scalloped form may increase the strength of the interface between the two types of tissue.[69,71] The scalloping has been observed to be larger in posterior teeth that are exposed to heavier occlusal wear.[72] Some collagen fibers from the dentinal material extend through this transitional area and are embedded in the more highly mineralized enamel. These embedded collagen fibers may also add to the overall strength and resilience of the junction between the two tissues.[73] A 200- to 300-µm layer of dentin that transitions from the bulk dentin into the DEJ complex, which includes the mantle layer of dentin, has been described as a *soft zone* of dentin because it contains tubules with little to none of the highly mineralized peritubular lining seen in other sections of dentinal tubules. This soft zone exhibits a reduced stiffness in comparison with bulk dentin, and it may play a significant role in providing a cushioning soft layer between the enamel and bulk dentin of the tooth.[74]

Support

Tooth strength, rigidity, and integrity rely on an intact dentinal substrate. To appreciate the magnitude of occlusal loading, a mean maximum bite force of 738 N (166 lb)[75] applied to an average contact area of 4 mm^2 distributed over 20 occlusal contacts[76] produces more than 26,000 psi (180 MPa) of stress. Investigators have

reported that resistance to tooth fracture is compromised with increasing depth and/or width of cavity preparation.[77,78] Dentin has a tensile strength of 40 MPa (6,000 psi) and a compressive strength of 266 MPa (40,000 psi).[79] A posterior tooth with an endodontic access preparation retains only a third of the fracture resistance of an intact tooth.[80] In vitro studies report that large mesio-occlusodistal (MOD) preparations increase the strain or deflection of facial cusps threefold compared with that of intact control teeth, and coronal stiffness decreases more than 60%.[80] Elastic deformation of the crown and cuspal flexure are factors that can contribute to noncarious cervical lesions,[81] cervical debonding of restorations,[82] marginal breakdown,[83] fatigue failure, crack propagation, and fracture.[84,85] Removal and replacement of dental restorations over a patient's lifetime generally result in successively larger or deeper preparations.[86] Therefore, to preserve coronal integrity, a conservative approach that combines localized removal of carious tooth structure with preservation of sound tooth structure, placement of sealants, and placement of bonded restorations is recommended.[87] If a large preparation is required, the dentist should consider complete coverage of the occlusal surface with an onlay or a crown. As with enamel, there is evidence that chemical treatments applied to the tooth during certain dental treatments, such as higher concentrations of tooth-whitening agents, may have adverse effects on the fracture toughness of dentin.[60,88]

Morphology

Dentin is primarily composed of small, thin apatite crystal flakes embedded in a protein matrix of cross-linked collagen fibrils. The thickness of the apatite crystals varies from 3.5 nm near the DEJ to about 2 nm close to the pulp. Although a random orientation is found for most of these thin apatite crystal platelets, it has been observed that there is a significant increase in the amount of parallel coalignment of these particles in areas where high strain might be expected, such as cusps.[89] The odontoblast, with its cell body at the pulp periphery and its extended process within the dentinal tubule, secretes the organic dentin matrix and regulates mineralization. The converging paths of the odontoblastic processes form channels or tubules traversing the full 3.0- to 3.5-mm (3,000- to 3,500-μm) thickness of the dentin from the pulp to the DEJ. The mean tubule diameter near the pulpal wall is 2.5 μm. Within the first 0.5 mm from the pulp, the mean diameter decreases rapidly down to 1.9 μm, tapers more gradually over the next 2 mm of its length, and then tapers very little in the final 1.5 mm of the tubule to terminate in a diameter of 0.8 μm at the DEJ.[90] The

tubules comprise about 10% of dentin volume.[90] Near the axial coronal area of the DEJ, the tubule paths form a double curve or S shape, whereas tubules near the DEJ in occlusal areas and root surfaces form a relatively straight path to the pulpal interface. In mature dentin, the odontoblastic process extends within the dentinal tubule to about one-third the dentinal thickness.[91] Variations, such as the density of tubules and the degree and quality of cross-linking of collagen fibers, have been observed between the structure of coronal dentin and root dentin. These variations could affect the degree of demineralization achieved with phosphoric acid or other acids, the stability and durability of the hybrid layer, and bond strength in these areas of the tooth.[92] (See chapter 9 for more on bonding to dentin and enamel.)

Unlike enamel, which is acellular and predominantly mineralized, dentin is, by volume, 45% to 50% inorganic apatite crystals, about 30% organic matrix, and about 25% water. Dentin is typically pale yellow in color and is slightly harder than bone. Two main types of dentin are present: *(1) intertubular dentin*, the structural component of the hydroxyapatite-embedded collagen matrix forming the bulk of dentin structure, and *(2) peritubular dentin*, limited to the lining of the tubule walls (Fig 1-13). Peritubular dentin has little organic matrix but is densely packed with miniscule apatite crystals. Though primary intertubular dentin remains dimensionally stable, the hypermineralized peritubular lining gradually increases in width over time.[93] The relative and changing proportions of mineralized crystals, organic collagen matrix, and cellular and fluid-filled tubular volume determine the clinical and biologic responses of dentin. These component ratios vary according to location (depth) in the dentin, age, and the history of trauma to the tooth.

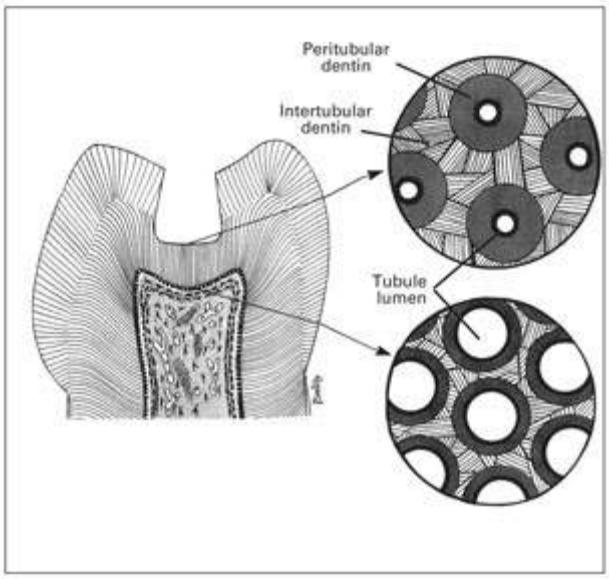


Fig 1-13 Dentin near the DEJ *(top)* and near the pulp *(bottom)* is compared to show relative differences in intertubular and peritubular dentin and in lumen spacing and volume.

Permeability

Although functional in forming and maintaining dentin, the open tubular channels of dentin compromise its function as a protective barrier. When the external covering of enamel or cementum is removed from dentin through cavity preparation, root planing, caries, trauma, or abrasion and erosion, the exposed tubules, if patent, become conduits between the pulp and the external oral environment. The exposure of the tubules with cavity preparation is somewhat offset by a layer of tenacious grinding debris, the *smear layer*, which adheres to the surface and partially plugs the tubular orifices.[94] For optimum success, dentin bonding systems must remove, modify, or penetrate this organic-inorganic barrier to facilitate resin diffusion and micromechanical bonding with the demineralized dentinal substrate.[95] However, the removal of the smear layer with acids during dentin bonding procedures causes an increase in the local dentinal permeability along with an outward fluid movement from the tubules, which results in adverse conditions for adhesive bonding.[96] Currently available dentin adhesives are capable of establishing an effective immediate bond strength in the wet environment presented by the cut dentinal surface; however, there are still some questions concerning the long-term effectiveness of these bonds because of problems involving the degradation of the resin and/or hydrolysis of the collagen.[97–99]

When injury or active caries affects dentin, the immediate inflammatory response is pulpal vasodilation, increased blood flow, and increased interstitial fluid pressure, which results in an increased outward flow rate of tubular fluid.[100] In vitro studies have shown that the fluid outflow may partially counteract the inward diffusion of toxic solutes through the tubules by 50% to 60%.[101] In addition, vasodilation and temporary gaps between the junctional complexes of adjacent odontoblast cells accommodate the passage of plasma proteins, such as albumin and immunoglobulins, into the dentinal fluid. These components agglutinate within the tubules to limit the diffusion to the pulp of exogenous stimuli and possibly to provide a direct immune response to bacteria.[102,103] Thus, with exposure of the tubules, a vascular response and accelerated outward flow of the tubular fluid constitute an immediate protective response. Nonetheless, tubules that are blocked or constricted provide the pulp with better protection from the permeation of noxious substances.

The diffusion gradient is reduced by both smaller tubular diameters and greater tubular lengths, ie, greater remaining dentinal thickness (RDT). Indeed, the functional diameter of the tubule is only a fraction of the anatomical lumen, because intratubular cellular, collagenous, and mineral inclusions restrict flow through the

tubular channels.[104] Furthermore, the length of tubules and the inherent buffering capacity of a full thickness of dentin create an effective biofilter of diffusion products.[105,106]

There are also regional differences in dentinal permeability. The coronal occlusal dentin (pulpal floor of a cavity preparation) is inherently less permeable than is the dentin around the pulp horns or axial surfaces.[107,108] As a result, although the fissured occlusal surfaces of posterior teeth often require cavity preparation, only about 30% of the subjacent dentinal tubules are patent over their entire length. However, gingival areas of preparations, such as prepared proximal boxes or crown margins, which are relatively more susceptible to microleakage and development of recurrent caries lesions, are located where the dentin is most permeable.[109,110]

The presence of bacteria or their by-products in deep dentin causes an acute histopathologic and inflammatory response within the pulp.[111,112] Even restored teeth are at risk of continued toxic diffusion through the phenomenon of *microleakage*, the flux of substances between the oral environment and the restoration-tooth interface due to the presence of interfacial gaps and possibly the differing coefficients of thermal expansion of tooth structure and restorative materials[111] (Fig 1-14). No restorative material or technique can ensure a complete hermetic seal of the restoration-tooth interface, and leakage at the gingival (cementum or dentin) margins of resin-bonded restorations is commonly reported.[113,114] Through marginal defects, differential thermal expansion, and capillary action, various cytotoxic components or bacterial endotoxins may diffuse through the dentinal substrate to reach the pulp. Clinically, an open margin or leaking restoration contributes to a wide range of problems, from marginal stains to sensitivity and chronic pulpitis, and is therefore frequently cited as the reason for replacement of an existing restoration[115] (Fig 1-15).

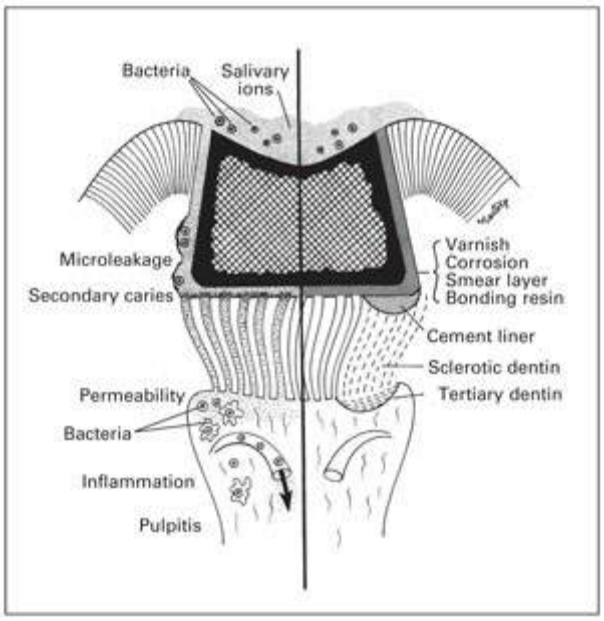


Fig 1-14 Leaking restoration interface *(left)*; sealed restoration interface *(right)*. Microleakage is exacerbated by polymerization shrinkage, condensation gaps around the restorative material, and/or differences in thermal expansion. When microleakage is present, the tubule openings in dentin form a potential pathway between the oral environment and the pulp. Various restorative materials, together with the tooth's defenses of tubule sclerosis and reparative dentin, restrict the noxious infiltration.

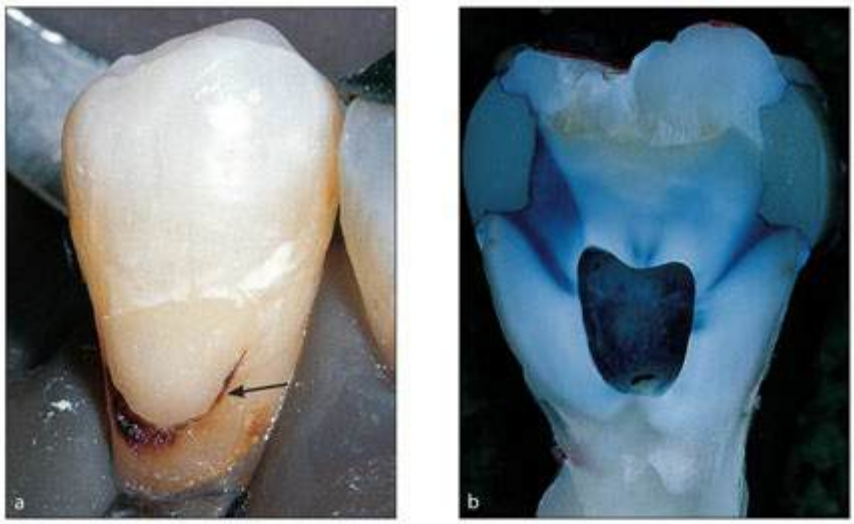


Fig 1-15 *(a)* Failed resin composite restoration. Polymerization shrinkage and cervical debonding created a restoration-wall gap defect (*arrow*, cervical margin), leading to microleakage and secondary caries. *(b)* In vitro dye penetration reveals microleakage and diffusion through dentinal tubules.

Tubular conduits connecting the pulp to the external oral environment create a virtual micropulpal exposure. Newly erupted teeth with relatively open tubules are particularly vulnerable to pulpal effects from active caries and rapid penetration of

bacteria.[116] Without treatment, loss of tooth structure due to carious demineralization or excessive wear results in a diminished thickness of dentin separating the pulp from the oral environment. If the threatening stimuli are moderate and slow in developing, the dentin-pulp complex may have time to hypermineralize or sclerose the tubular channels or to add new tertiary dentin at the pulp-dentin junction (PDJ). Blockage of the tubules and dentinal repair are the most important defensive reactions of the dentin. However, with trauma, rapid advance of a caries lesion, or deep cavity preparation, a minimal RDT with numerous open tubules renders the pulp vulnerable to the influx of noxious substances. Without intervention, bacteria eventually reach the level of the PDJ, and pulpal necrosis is the probable outcome.[117]

Dentinal substrates

The form and constituency of dentinal tissue are not static, and changes in its basic components may occur throughout the life of the tooth. Some of the variations in the dentinal tissue occur during its natural sequence of development or aging, while other variations result from the effects of external factors on the tooth, such as caries, injury, or wear. An understanding of these variations in the dentinal tissue is important because they are related to the long-term success of dental procedures and therapies.

Primary and secondary physiologic dentin

Bioactive signaling molecules and growth factors in the inner dental epithelium differentiate ectomesenchymal cells of the dental papilla into mature odontoblast cells. They synthesize and secrete extracellular organic matrix, which, following mineralization, forms the primary and secondary physiologic dentin[93,118] (Fig 1-16). The first-formed, 150-µm-thick layer of primary dentin subjacent to the enamel is termed *mantle dentin*. It differs from other primary dentin in that it is 4% less mineralized, and the collagen fiber orientation is perpendicular rather than parallel to the DEJ. Following mantle deposition, odontoblasts begin to form odontoblastic processes and create tubules as the cell bodies converge pulpally. When mature, as long as the root apex remains undeveloped and open, the odontoblasts produce primary dentin, mainly intertubular dentin, at a rate of 4 to 8 µm/day. Approximately 2 to 3 years following tooth eruption, and coincident with root apexification, the bulk of dentin surrounding the pulp chamber and canal systems, termed *circumpulpal dentin*, is completely formed. The synthesis of dentin then slows to 1 to 2 µm/day,

decreasing in rate with age but continuing as long as the tooth is vital. The tubules remain regularly spaced and continuous with tubules within the primary dentin. As the tooth matures, this secondary dentin is distributed gradually and asymmetrically to create pulp-chamber volume reduction with a relatively constricted occlusogingival dimension. The pulp horns and root canals are also gradually reduced in volume. With this ongoing reduction in size of the pulp, the inherent risk of pulpal exposure during cavity preparation tends to decrease with the age of the patient. Before starting a cavity preparation or crown preparation, the dentist should radiographically assess the size and location of the pulpal tissues in relation to the size and location of the caries lesion in order to anticipate the need for an indirect pulp capping procedure and to attempt to avoid a pulpal exposure.



Fig 1-16 Primary and secondary dentin. *(left)* Primary dentin and large pulp chamber and root canals are shown in a mandibular molar after eruption but before completion of root formation, when accelerated primary dentin formation ceases and secondary dentinogenesis begins. *(right)* Mature molar that has had gradual and continued deposition of secondary dentin. Note the large mesiobuccal pulp horn that is susceptible to exposure with deep cavity preparation. There is asymmetric deposition of secondary dentin on the pulp chamber roof and floor to narrow the vertical dimension. (Courtesy of James A. Gillis, San Antonio, Texas.)

Outer dentin

In the first-formed dentin near the DEJ, the tubules of the outer dentin (Fig 1-17; see also Fig 1-13) are relatively far apart, and, with time, mineral supplementation of peritubular walls progressively narrows the lumen. With relatively fewer tubules at the periphery, around 20,000 tubules/mm^2, and small tubule diameters of approximately 0.8 µm, the tubule lumens only constitute about 4% of the surface area of cut outer dentin[119] (see Fig 1-17). However, there is extensive terminal branching of the tubules in the outer 250 µm of dentin and regularly spaced connecting branches between tubules. Smaller fine canaliculi and even microfine 0.1-µm pores extend from the tubule walls to permeate the intertubular dentin (Figs 1-18 and 1-

19). Similar to the vascular system, this highly interconnected and fluid-filled tubular system acts as a transporting medium for mineral exchange and for bioactive molecules released from the dentinal matrix.[90,120] This networking of tubules may account for the paradox that pressure as localized as an explorer tip moving across a surface of cut dentin may indirectly stimulate a plexus of neurons to cause a sensation of pain. Also, when the prepared dentinal surface is acid etched for resin bonding, the highly mineralized peritubular walls are the first to be solubilized to create wide funnel-shaped tubules and expose connecting branches. Resin penetration into tubules and branches, together with the micromechanical bond of the resin-dentin hybrid layer, form a mechanical interlocking of resin tags to create the best possible bond to the etched dentin substrate.[121]

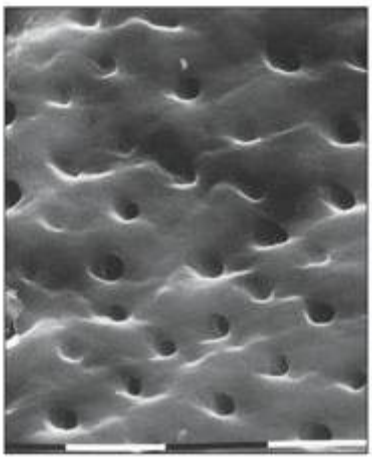


Fig 1-17 Scanning electron photomicrograph of tubules in outer dentin. All highly mineralized peritubular dentin has been removed in the specimen preparation. (Bar = 10 µm.)

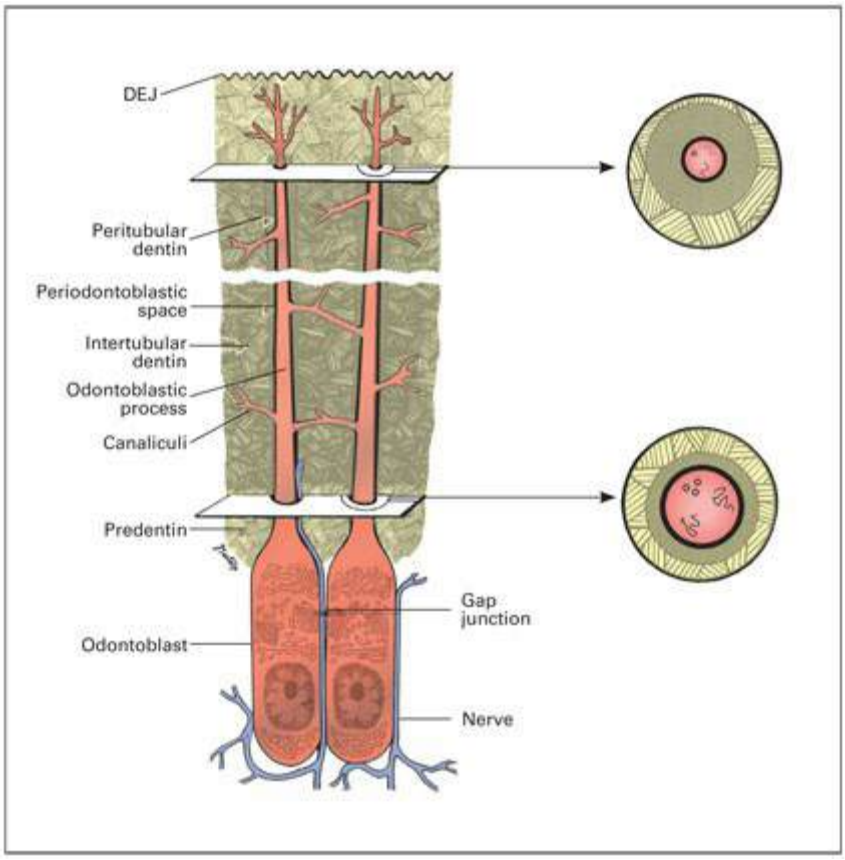


Fig 1-18 Odontoblastic cell process and tubule system through dentin. Continual deposition of peritu-bular dentin and minerals, accelerated by a chronic, noxious stimulus, gradually occludes the tubules peripherally. Note the terminal branching and interconnections between odontoblastic cell processes and between cellular walls. Direct neural penetration of dentin is limited to less than 20% of the tubules and rarely beyond the predentin.

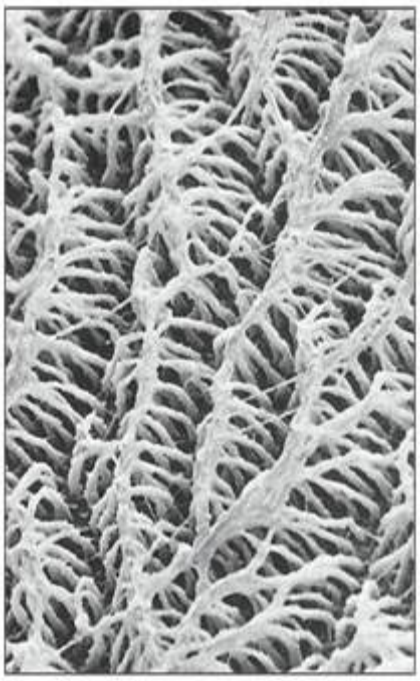


Fig 1-19 Scanning electron photomicrograph of resin penetration into a dentinal tubule system after etching with phosphoric acid. The dentin was then demineralized with hydrochloric acid and the organic component removed with sodium hypochlorite, leaving the resin to illustrate tubule configuration. Note the cross-branching of tubules; this illustrates the complexity of the dentinal tubule. (Courtesy of Jorge Per-digao, Minneapolis, Minnesota.)

It has been shown that the resin bond to dentin deteriorates over time.[97,98,122] The degradation of denuded collagen in the layer of resin-infiltrated dentin is currently considered to be one of the main problems related to the deterioration of this bond. Current research attempting to identify the cause of this degradation of the bond between resin and dentin suggests a link to water sorption–induced hydrolysis of the hydrophilic resin components[123] and the breakdown of the protein within the collagen by a group of enzymes known as *matrix metalloproteinases* (MMPs).[124] MMPs are released from the dentinal matrix during the acid-conditioning step and degrade the strength of the bond by attacking the collagen fibers in the hybrid layer.[125] Evidence has been presented for various suggested approaches to resolve this bond degradation problem. One approach suggests that the use of an MMP inhibitor, such as chlorhexidine digluconate (often used during dental procedures as an antibacterial agent) or galardin (a synthetic protease inhibitor), incorporated into dentin bonding systems may be able to counterattack the effects of this endogenous enzyme on the dentin bond.[125,126] Another approach suggests the incorporation of some form of amorphous calcium phosphate nanoprecursor into the dentin bonding system to attempt to accomplish both interfibrillar and intrafibrillar remineralization of the degraded dentinal tissue.[127] Adhesion to tooth structure is discussed in more detail in chapter 9.

Because the processes of the odontoblastic cells extend no farther than the inner third of adult dentin (approximately 1.0 mm), cavity preparations or caries lesions confined to the outer dentin do not directly sever or degrade the vital cellular component of the dentin-pulp complex. Peripheral preparations or lesions with a substantial remaining dentin thickness of 2.0 mm or more provide a sufficient physiologic barrier to safeguard pulpal health from routine restorative techniques.[128] One important exception is an extensive crown preparation without water coolant and with constant, as opposed to intermittent, cutting, which may generate a level of heat or rate of temperature increase capable of creating histopathologic evidence of pulpal injury.[129]

Inner dentin

The dentinal substrate near the predentin and PDJ is quite different from that near the DEJ. The 20-µm-thick predentin layer consists of newly secreted organic matrix awaiting mineralization. The converging tubules at the predentin, the portion of the dentin closest to the pulp, number up to 58,000/mm^2 in cross section and contain the processes of the odontoblast cells[130] (Figs 1-20 and 1-21). Careful cavity preparation and proper restorative technique are required to limit surgical trauma to the odontoblast cell bodies and prevent their injurious displacement into the tubules,[131] but with good technique and a healthy pulp prior to tooth preparation, the likely outcome of a deep preparation is pulpal healing without clinical symptoms. The tubule diameters near the PDJ are larger (2.5 to 3.0 µm), the distance between tubule centers is half that between tubule centers at the DEJ, and the peritubular dentin is diminished in thickness or absent.[132] At the PDJ, the area of the intertubular dentin is as little as 12% of the surface, and the volume of the fluid-filled tubule lumens approaches 80%.[90,133] Therefore, at this level, the dentin is more permeable and about 22 times wetter than the dentin at the DEJ.[134] The fluid in the dentinal tubules is an extension of the interstitial fluid within the pulp, which has a positive pressure of 5 to 20 mm Hg. Therefore, the deeper the cavity preparation, the greater the outward flow of dentinal fluid from the exposed tubules to "wet" the cut surface. Some moisture has been shown to facilitate dentin bonding.[135] However, studies of various bonding systems incorporating simulated pulpal pressure in deep dentin have demonstrated "overwet" conditions and lower bond strengths for some adhesives, but not for others.[91,136–138] Also, deep cavity preparation extending near to the pulp may injure the cellular tissues, and a minimal RDT, whether from preparation, trauma, or a caries lesion, places the pulp in close proximity to toxic or immunologic stimuli.[139]

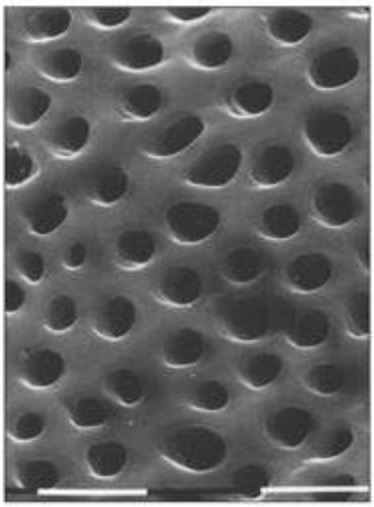


Fig 1-20 Scanning electron photomicrograph of tubules in inner dentin of tooth in Fig 1-17. The

section was approximately parallel to the walls of the pulp chamber. All highly mineralized peritubular dentin has been removed in the specimen preparation. (Bar = 10 µm.)

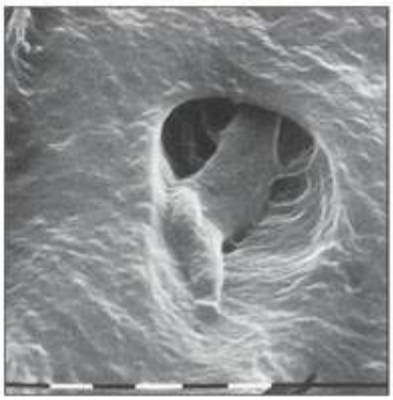


Fig 1-21 Magnified tubule orifice with collagen matrix and odontoblastic process. (Bar = 1.0 µm.)

Carious dentin

The caries process is driven by the presence of a biofilm containing acid-producing bacteria on the tooth surface. Without intervention, a progression of destructive changes occurs, prompting pulpal and dentinal responses. This begins with the subsurface enamel lesion and is followed by dentin demineralization, cavitation, infection of demineralized dentin, dentin-matrix dissolution, and, ultimately, pulpal necrosis.[140] The degree and type of response is related to the caries activity, which may vary from active and rapidly progressive to chronic and slowly progressive or arrested. Over time, with the changing interplay of the oral environment, lesion development, host response, and preventive practices, the same lesion may assume any of these forms.

The earliest dentinal response occurs adjacent to the center or apex of the enamel lesion, where the deepest demineralization of enamel rods approaches the DEJ. Even before the enamel lesion reaches the DEJ histologically, acid protons, solubilized matrix components, and released bioactive molecules diffuse through the tubules contiguous with the affected enamel rods to stimulate morphologic changes and metabolic activity in the affected primary odontoblasts. Mineralizing components are released into the tubular fluid of the periodontoblastic space to augment the existing peritubular walls and to form a localized zone of hypermineralized dentin subjacent to the enamel lesion of permeable enamel rods.[141,142]

When the enamel rod dissolution and enamel porosity reach the DEJ, the hypermineralized zone then becomes subject to accelerated acid dissolution. In

contrast to enamel, dentin demineralization is more rapid because of the tubular network and the high surface-to-volume ratio of the small hydroxyapatite crystallites embedded in the collagen.[143] Clinically, the affected dentin is often distinguished from normal dentin by decreased hardness and by a yellow-brown discoloration due to acid effect on the organic dentin matrix or possibly from exogenous staining.[144] Unchecked, demineralization of the soft and discolored dentin progresses toward the pulp (Fig 1-22). However, as long as the enamel surface remains intact, the dentin lesion is relatively sterile and devoid of viable bacteria.[16] Incipient, noncavitated lesions may be arrested with plaque control or with other noninvasive preventive therapies.[145]

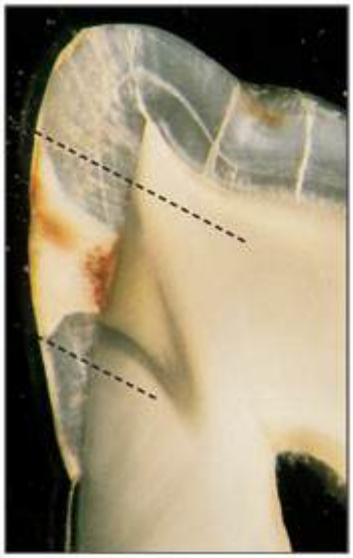


Fig 1-22 Precavitated smooth-surface caries lesion with demineralization of enamel rods in the center of an enamel lesion extending to the DEJ; dentin demineralization is guided by the direction of dentinal tubules. Note that the affected dentin area at the DEJ is adjacent to the enamel rods that have suffered demineralization at the outer surface *(dotted lines)* as well as those that have been demineralized to the DEJ. (Reprinted from Bjørndal and Mjör[140] with permission.)

A pivotal point of caries lesion progression occurs if the 20- to 50-µm surface layer of enamel over the internal enamel lesion fractures so that the surface becomes cavitated. Within the defect, which is generally inaccessible to brushing or flossing, the facultative cariogenic bacteria multiply to generate a destructive acidic environment. A pathologic cycle of tooth destruction, infection, and tubular invasion of the dentin structure ensues (Fig 1-23; see also Fig 1-5). Following the demineralization of the peritubular walls and the intertubular crystals, proteolytic enzymes from the bacteria disrupt the cross-linked collagen framework of intertubular dentin. Clinically, advanced or acutely infected dentin differs from normal dentin or dentin of an arrested caries lesion in that it is soft, readily

excavated, wet, and generally light yellow to orange in color.[146] This amorphous lesion is referred to as *infected dentin* and histologically as the *zone of destruction*. Beneath this zone, where the dentin matrix is still intact and limited bacterial penetration is confined to the tubules, the dentin is termed *affected dentin*.[147] Only select microorganisms invade the tubules. In the outer superficial or cavitated lesion, strains of gram-positive streptococcus prevail, whereas anaerobic rods are the bacteria primarily found in deep dentinal tubules and infected pulps and root canals.[148] If the caries lesion progress is gradual, the pH gradient in the deeper tubules below the affected dentin promotes recrystallization of the solubilized minerals. The precipitated crystals, called the *zone of sclerosis*, occlude the tubule lumens to restrict diffusion of toxins to the pulp. However, without operative intervention, the acidic and bacterial front eventually breaches the hard tissue defenses that protect the pulp. The infusion of endotoxins and bacterial antigens into the pulp evokes severe immunologic, inflammatory, and cellular responses, with the probable outcome of irreversible pulpitis. However, several clinical studies of sealants, sealed restorations, and indirect pulp capping procedures in vital teeth suggest that even advanced or infected dentin caries lesions may be arrested in situ if the bonded materials successfully seal the lesion and entomb the bacteria.[149,150]

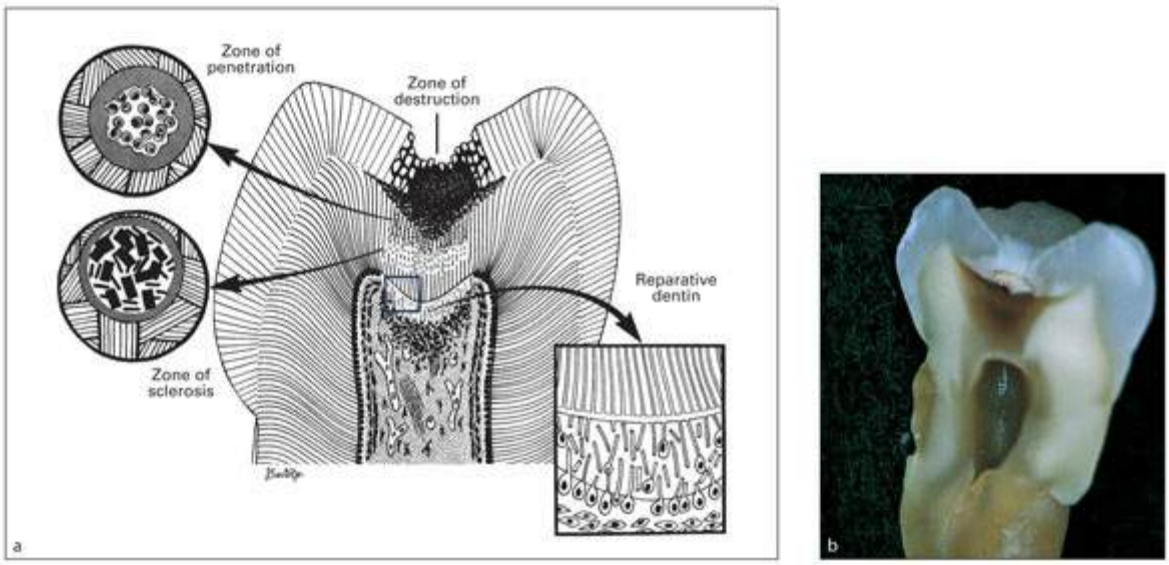


Fig 1-23 Tooth response to carious destruction of tooth surface. Acid demineralization and enzymatic destruction of the collagen matrix lead to cavitation, an irreversible change. *(a)* Bacteria fill and demineralize the lumens of the tubules peripherally, but dissolved minerals reprecipitate at a deeper level to augment sclerosis and hypermineralization of subcarious dentin. Reparative dentin with irregular and noncontinuous tubules forms a final barricade against bacterial metabolites. *(b)* Note the lateral spread of the caries lesion at the DEJ and a hypermineralized sclerotic zone around the pulp.

In the event of cavitation and dentinal infection, restorative treatment is necessary to remove the infected dentin and restore the integrity of the coronal surface. Discoloration is an unreliable guide to excavation of carious dentin, but the degree of dentin hardness, as determined by tactile feedback from excavating burs and hand instruments, is the most reliable guide to differentiation between infected, affected, and normal dentin.[151] As with dentin of different depths, some types of bonding systems are better suited to dentin altered by caries. Etch-and-rinse systems have bonded well to moist, caries-affected dentin,[152] and extended etching times have helped.[153] However, self-etching systems provide significantly reduced bond strengths when applied to affected and infected dentin.[154,155]

Altered dentin

Dentin experiences alterations in its morphology due to both the aging process and the localized defensive and repair responses to injury resulting from caries or wear. The mechanisms of biologic control and coordination of these pulpal and dentinal responses are beginning to be understood. The hard tissue responses to injury include tubular hypermineralization and sclerosis, which restrict the tubular diffusion of noxious agents. Also, tertiary dentinogenesis adds new barrier dentin at the pulpal interface. At the same time, a pulpal response is underway, including activation of odontoblastic and subodontoblastic cells, proliferation of vascular and neural tissues that support these cells, a heightened immune response, and inflammation.

Sclerotic dentin. A combination of abrasion, attrition, erosion from dietary sources or gastric acid, or occlusal stress may lead to the loss of enamel, cementum, and dentin. The progressive loss of tooth structure at the CEJ typically presents as a wedge-shaped defect. Although the etiology of the lesion is multifactorial, it is not primarily a result of demineralization from bacteria-produced acids, and it is termed a *noncarious cervical lesion* (NCCL). The exposed radicular dentin of an advanced NCCL differs in appearance from cut coronal dentin in that it is generally deep yellow in color and has a transparent, glossy surface[156] (Fig 1-24). These lesions may be episodically and acutely sensitive to touch or to changes in temperature, so the dentin is termed *hypersensitive*. The condition is directly related to the percentage of open or patent tubules between the exposed root surface and the pulp; nearly 75% are present in sensitive dentin versus about 24% in insensitive root dentin.[157] Sclerotic dentin is characterized by hypermineralization or blockage of the

tubules with whitlockite crystals and by a denatured collagen network. Studies have demonstrated acid-resistant, bacteria-embedded, hypermineralized, layered plaques on the surface of the sclerotic dentin of NCCLs.[158] The altered surface and substrate limit the formation of both the hybrid layer and the resin tags, so that the bond strengths observed during in vitro testing to sclerotic dentin in a NCCL is 20% to 40% less than the bond strengths to dentin in artificially created wedge-shaped lesions prepared with carbide cutting burs in the cervical area.[159] However, in a recent 8-year clinical study involving the preparation and restoration of 112 NCCLs, no increase was observed in the loss rate of restorations placed in sclerotic dentin when compared with restorations placed in normal dentin.[160] The optimum mechanical or chemical preparation of the hypermineralized surface, types of bonding systems, and techniques used to bond restorations to sclerotic dentin are still being investigated.[156,161]



Fig 1-24 Large NCCLs in the maxillary left central incisor, canine, and first premolar that illustrate the appearance of sclerotic dentin.

Hypermineralized dentin. Following root formation and for as long as the tooth is vital, the odontoblasts slowly produce extracellular dentin matrix and concentrate minerals for the production of physiologic secondary dentin. It is theorized that a portion of the mineralizing components released into the tubules gradually augments the thickness of the mineralized peritubular walls. As the tooth matures, beginning at the periphery of the dentin, the tubules become progressively hypermineralized and, with constriction of the lumen, less permeable. Just as secondary dentin deposition is primarily physiologic, tubular hypermineralization or sclerosis is an age-related process of the coronal dentin and, especially, of the root dentin.[162] However, with an external stimulus or irritation, such as a slowly progressing caries lesion, attrition, or restorative procedures, the rate of mineral augmentation to the tubular walls can be accelerated.

Mineral crystallization within the tubules, as in sclerotic dentin in the walls of

NCCLs, is also an important defensive response to attrition and to an active caries process in dentin. The apatite minerals of the inorganic dentin are dissolved in the acidic environment of the peripheral cavitated lesion. The supersaturated acidic solution is diluted and buffered by diffusion through and contact with the tubular walls, and the pH kinetics reverse to favor reprecipitation of calcium and phosphates. Platelike, cuboidal, or rhomboid mineral crystals form to barricade the open lumen[163] (Fig 1-25). Similar crystals are observed within tubules of coronal or root dentin exposed to the oral environment through attrition or abrasion.[164]

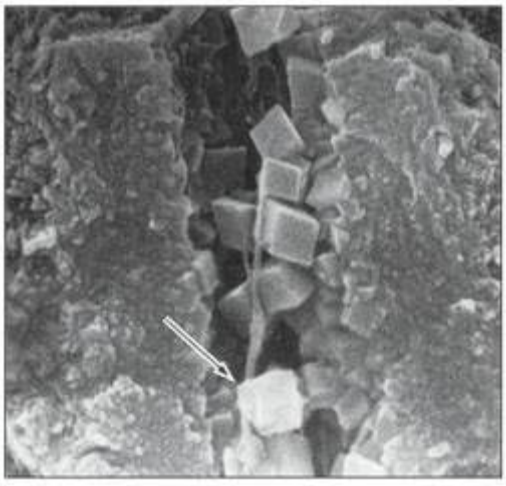


Fig 1-25 Intertubular precipitation crystallites *(arrow)* nearly occluding the dentin tubule. (Original magnification ×27,000. Reprinted from Yoshiyama et al[157] with permission.)

The combination of peritubular wall thickness and intratubular crystals creates a zone of hypermineralized dentin beneath exposed or carious dentin, the zone of sclerosis or *translucent zone* (see Fig 1-23a). Sclerotic dentin is frequently found beneath both active caries lesions and restorations and is an important defense reaction of the hard tissues because it limits the permeability of dentin.[165] Rate of caries lesion progression and patient age are important factors. Rapidly progressive caries lesions in newly erupted teeth can lead to dead tracts, empty tubules in which the odontoblast and its process are destroyed before any defensive restriction of the tubules can occur.[166] In one study, the sclerotic zones beneath caries lesions in young adults limited the dentin permeability to only 14% of the permeability of noncarious controls. With subjects aged 45 to 69 years, the dentin beneath caries lesions was completely impermeable.[167] However, as with the sclerotic dentin of NCCLs, the altered sclerosed dentinal substrate may limit bond strengths of restorative systems. It is not clear to what extent the genesis of the sclerotic dentin is purely physicochemical or biologically controlled. However, hard tissue responses to external noxious stimuli generally occur in conjunction with an active biologic

cellular response of tertiary dentinogenesis.

Tertiary dentin. Newly formed dentin at the dentin-pulp interface compensates for the loss of peripheral dentin from caries or injury and may provide a superior pulpal seal against noxious diffusion through the tubules (Fig 1-26). If the stimulus is relatively low-grade, such as from an incipient enamel caries lesion, the primary odontoblasts are metabolically reactivated to produce a localized tertiary dentin termed *reactionary dentin* (Fig 1-27a). At the same time, complex biochemical signaling systems promote proliferation of supportive vascular and neural tissues among the affected odontoblast cells. Paradoxically, despite being reactivated, the odontoblasts do not assume the increased size and complexity observed during secretion of the primary dentin matrix.[118] The affected odontoblasts are smaller and have a decreased cytoplasm-to-nucleus ratio.[170,171] The rate of reactionary dentin formation, the inclusion of tubular and cellular components, and the continuity of tubules in reactionary dentin with tubules in the overlying dentin all vary according to the severity of the stimulus. Reactionary dentin resulting from a mild, slowly progressing caries lesion may resemble secondary dentin with connecting tubules between the two tissues. An active noncavitated lesion may result in atubular reactionary dentin.[172] With a rapidly progressing caries lesion, neither tubular hypermineralization nor reactionary dentin has an opportunity to form.[173]

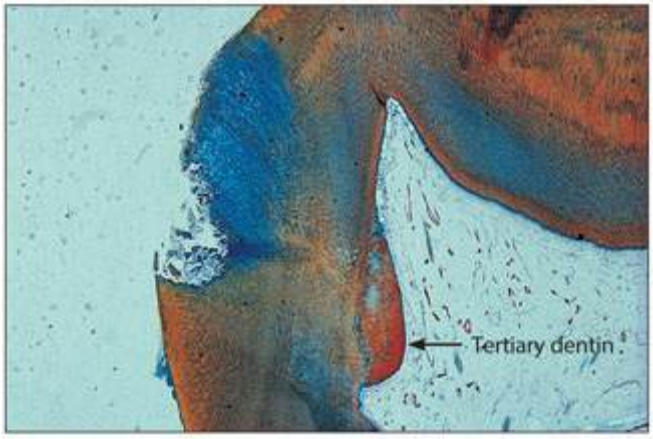


Fig 1-26 Reparative dentin forming a localized defense barrier and replacement for lost carious dentin at the periphery. The effectiveness of the diffusion barrier is shown by the absence of pulpal inflammation. (Reprinted from Trow-bridge[168] with permission.)

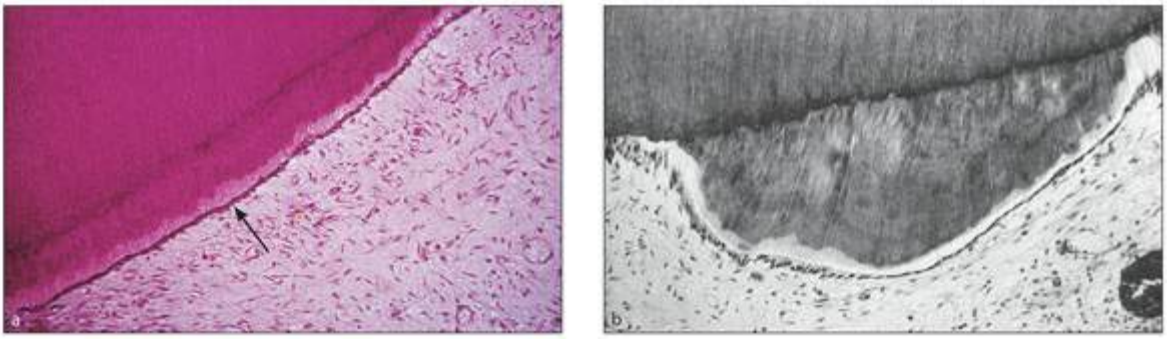


Fig 1-27*(a)* Reactionary dentin. Note the layer of intact primary odontoblast cells *(arrow)* and the regular pattern of tubules continuous with those of the physiologic secondary dentin. (Hematoxylin-eosin stain, original magnification ×56.) (Reprinted from Trowbridge[168] with permission.) *(b)* Reparative dentin. The dentin is produced by differentiation of subodontoblastic cells that replace the primary odontoblasts killed from the effects of caries. The tubules are less regular and not continuous with those of the overlying dentin. (Reprinted from Trowbridge[169] with permission.)

When cavity preparation reduces the RDT to less than 0.25 mm, the trauma to the odontoblastic cell bodies and their processes reduces their numbers by up to 41%.[174] Even before restorative tooth preparation, an advancing caries lesion generates hydrogen ions and endotoxins that have the potential to critically injure the primary odontoblasts. With a mechanical pulpal exposure, 100% of the primary odontoblasts are destroyed. Yet, with vital pulp tissue surrounding the area of destroyed cells, the dentin-pulp complex can recruit other pulpal cells, including stem cells, to become odontoblast-like cells.[175] The damaged odontoblasts appear to be replaced by differentiating dental pulp cells that show an increased production of collagen type I, which provides the basic framework for dentin, along with dentin sialoprotein and phosphophoryn, both of which are proteins that control the deposition of apatite crystals in dentin.[176] The cascade of biologic signals to other pulpal cells is complex because recruitment, migration to the odontoblastic layer, differentiation, and a change in form and function must precede the secretion of new dentin matrix. The process takes 20 to 40 days before dentinogenesis begins.[177] With a mechanical pulpal exposure, the formation of a tertiary dentin "bridge" is expedited by direct pulp capping procedures that limit bacterial contamination and seal the exposure site.[178] Tertiary dentin produced by odontoblast-like replacement cells is termed *reparative dentin* (Figs 1-26 and 1-27b). The layer of reparative matrix formed at the dentin-pulp interface, termed *interface dentin*, is atubular and is therefore a superior barrier because it seals and terminates the tubular pathways of the overlying dentin.[179,180] Depending on the nature and rate of progression of the caries lesion, subsequent layers of tertiary dentin, equivalent to wound healing or scar formation, may vary in their tubular content.[172] Both reactionary and reparative

dentin can form beneath the same caries lesion because the periphery of a dentin lesion may be less advanced than the center.[171] To the extent that bacterial antigens and cytotoxins begin to breach the hard tissue barriers, the pulpal defenses must rely on increased vascular clearance, activated immune response, and inflammation.

Molecular signals link caries-inflicted or other injury to the tooth with the hard and soft tissue responses of the dentin-pulp complex. Until recently, mature dentin was considered inert and tertiary dentin a type of irritation response.[181] Recent research has identified molecules trapped within the mineralized dentin matrix that are equivalent to the signaling molecules responsible for the differentiation and activation of the embryonic preodontoblast cells. During primary dentinogenesis, they are synthesized and secreted by the odontoblasts into the extracellular dentin matrix. These bioactive molecules are embedded in both the soluble mineral and the insoluble collagen components of the dentin. Cells that have receptors for these growth factors, including odontoblasts and subodontoblastic cells,[182] are directed to various activities, including migration, metabolic functions, proliferation, and differentiation.[169]

Some of the active growth factors in dentin, such as bone morphogenetic protein-2 (BMP-2), are also present and instrumental in bone regeneration.[183] The foremost and most abundant of the numerous dentin growth factors is transforming growth factor-β1 (TGF-β1), one of a group of transforming growth factors.[184] With injury or caries, TGF-β1 is a pivotal molecule that initiates tertiary repair and modulates the inflammatory reaction.[185] Experimental extracts of solubilized dentin matrix contain the released growth factors. When placed in deep, atraumatically prepared cavities with a minimum RDT, the growth factors diffuse across the dentin to stimulate both reactionary and reparative tertiary dentin.[186] With caries, growth factors are released by demineralization, caused by either bacterial or restorative etching acids, and then diffuse pulpally through the dentinal tubules to interact with the odontoblasts and pulpal cells.[187,188] Also relevant to pulpal therapy, dentin extracts or chips of dentin experimentally placed on vital pulp tissue expedite reparative dentinogenesis.[189] The therapeutic application of growth factors incorporated into restorative materials to induce dentin impermeability and repair has been investigated to some extent. In an animal study, the inclusion of TFG-β1 in a calcium phosphate/hydrosoluble polymer composite pulp capping material was found to trigger stem cells in the pulp to differentiate into odontoblast-like cells and to induce the formation of tertiary dentin.[190]

Sensitivity

Dentin can be painfully sensitive, but there is no direct anatomical explanation. Although sensitive to thermal, tactile, chemical, and osmotic agents, dentin is neither vascularized nor innervated, except in about 20% of tubules that have nerve fibers penetrating inner dentin by no more than a few microns. The possibility of the odontoblastic cell body and process having a role in direct transmission of sensation is doubtful.[93] The odontoblastic process does not extend beyond the inner third of mature dentin.[191] In addition, the cell membranes of odontoblasts are nonconductive, and there is no synaptic connection between the odontoblastic cell and the contiguous terminal branches of the pulpal nerve plexus. Finally, pain sensation remains even when concentrated anesthetic solutions are placed on the dentin surface or when the odontoblastic layer is disrupted.[192] However, recent studies demonstrating the ability of odontoblasts to generate action potentials[193] have led some researchers to believe that odontoblasts could possibly be directly involved in the sensory transduction process of the tooth through some form of mechanosensory system.[194]

Brännström et al[195] proposed a theory based on the capillary flow dynamics of the fluid-filled dentinal tubules (Fig 1-28a). Tubular fluid flow of 4 to 6 mm/sec is produced by the application of a stimulus that expands or contracts the tubular fluid volume and/or creates rapid shifts in the rate and/or direction of the fluid flow. An outward flow apparently causes more sensation than does flow in a pulpal direction.[197] Common clinical procedures associated with tooth preparation, such as air drying, cold water rinses, or pressure from probing and cutting dentin, generate outward fluid displacement. The "current," or hydrostatic pressure, displaces the odontoblastic cell bodies and stretches the intertwined terminal branches of the nerve plexus to allow the entry of sodium to initiate depolarization and the perception of pain.[198] Adenosine triphosphate (ATP), released from damaged cells or by stimulation of endothelial cells, may provide a chemical rather than mechanical explanation for depolarization, because ATP receptors have been identified on terminal branches of the pulpal sensory neurons.[199]

Ahlquist et al[200] correlated intensity of pain with rapid changes of hydrostatic pressure applied to smear-free dentin in axial walls of cavity preparations. Other evidence supporting the hydrodynamic theory is the in vivo correlation of tubule patency with hypersensitivity of root dentin. With hypersensitive root dentin, the degree of tubule closure from intratubular crystals was directly correlated with decreased sensitivity to touch, air, or temperature stimuli.[158,201] Chemical

compounds that promote intratubular crystallization, such as oxalates or strontium chloride, are active ingredients in some toothpastes and professional desensitizing compounds.[202] Other materials, such as potassium nitrate, resin-glutaraldehyde mixtures, fluoride varnish, and resin bonding agents, have also been successfully utilized clinically to reduce dentin sensitivity by occluding dentinal tubules through various mechanisms.[203,204]

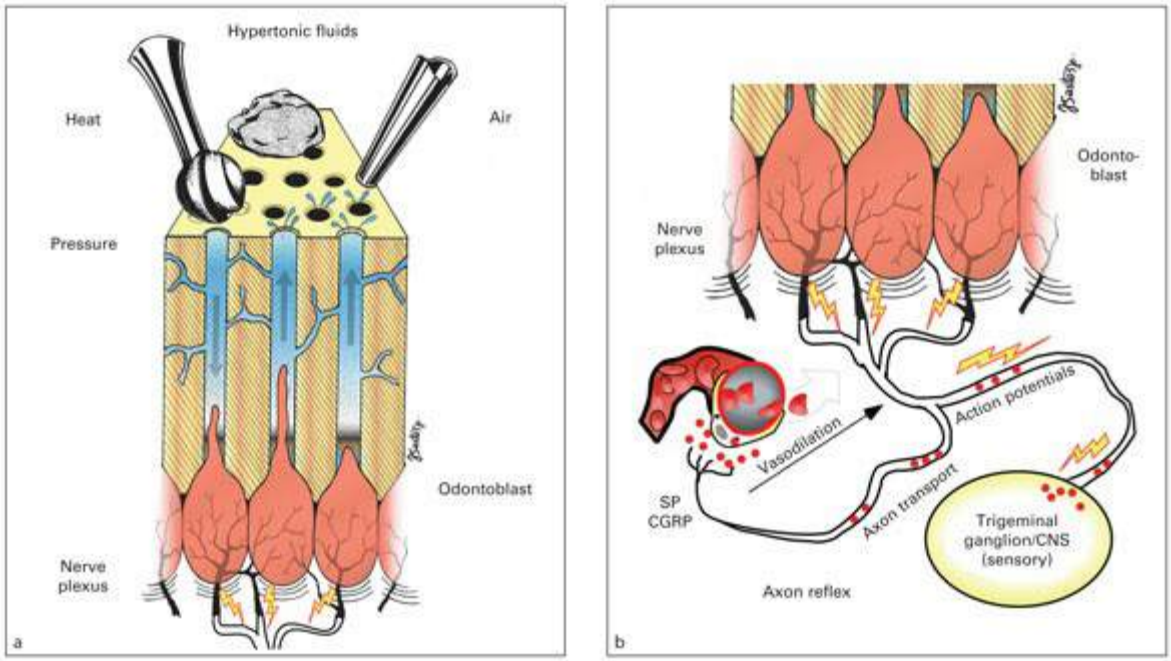


Fig 1-28*(a)* Hydrodynamic phenomenon explains the sensitivity of dentin, which is without significant innervation. Fluid dynamics of the tubules move the odontoblastic cell bodies and mechanically depolarize approximating sensory afferent nerve endings. *(b)* Axon loops from activated sensory nerves provide reflex efferent functions to pulpal, immunologic, and vascular cells by releasing neuropeptides, such as calcitonin gene-related peptide (CGRP) and substance P (SP), from terminal nerve endings. Whereas the sensory action potentials along nerve membranes are nearly instantaneous, the axonal transport of chemical factors and neuropeptide replacements may take hours or days.[196] CNS—central nervous system.

A painful stimulus-reflex response is protective as an alarm to avoid trauma or injury. However, the advantages of painful and sensitive dentin to foods and fluids are less clear, and pain is not a reliable indicator of histopathologic changes or of dentin demineralization caused by caries. It is possible that the major protective benefit of stimulating the sensory pulpal nerves is not the registering of pain in the central nervous system. Branches of these afferent nerves loop back via an axon reflex to stimulate the contractile components of the vascular complex (Fig 1-28b). When triggered, they release potent neuropeptides to activate vasodilation, increase blood flow, and elevate interstitial pressure.[196,205] Thus, rather than discomfort, homeostasis and pulpal defense may be the critical protective outcomes of the

hydrodynamic response.

Conclusion

The elastic nature of dentin provides stress relief for brittle enamel. The integrity of dentin is related to coronal strength and durability and can be compromised by carious demineralization, traumatic injury, wear, or poor restorative techniques. Unlike the relatively homogenous nature of enamel, the dentinal substrate varies considerably with location, age, and response to external stimuli. Differences in dentinal permeability, wetness and dryness, hypermineralization, and pathologic events complicate comparative research studies on dentin and dentin bonding systems. The dentinal matrix can no longer be considered biologically inert. Signaling molecules and growth factors are released in dentin injury or demineralization to activate tertiary dentin repair and to initiate mobilization of pulpal defenses. The barrier protection of the pulpal tissue is directly related to the impermeability of the dentinal tubules. When carious metabolites, toxins, and bacterial by-products access the pulpal tissues through these portals, both hard (dentinal) tissue and soft (pulpal) tissue defensive reactions are activated to seal the tubules and repair the damage. Because of the numerous interactions between tissues of the dentin and pulp, the two are often discussed together as one complex.

Pulp

Dental pulp,[93,206] composed of 75% water and 25% organic material, is a viscous connective tissue of collagen fibers and organic ground substance supporting the vital cellular, vascular, and nerve structures of the tooth. It is a unique connective tissue in that its vascularization is essentially channeled through one opening, the apical foramen at the root apex, and it is completely encased within relatively rigid dentinal walls. Therefore, it is without the advantage of an unlimited collateral blood supply or an expansion space for the swelling that accompanies the typical inflammatory response of tissue to injury. However, the protected and isolated position of the pulp belies the fact that it is a sensitive and resilient tissue with great potential for healing.

The dental pulp fulfills several functions:

- *Formative:* Generates primary, secondary, and tertiary dentin (dentinogenesis)
- *Nutritive:* Provides the vascular supply and ground substance transfer medium for

metabolic functions and maintenance of cells and organic matrix

- *Sensory:* Transmits afferent pain sensation (nociception)
- *Protective:* Coordinates inflammatory, antigenic, neurogenic, and dentinogenic responses to injury and noxious stimuli

Additional protective functions include homeostasis and clearance of noxious and antigenic substances through the vascular and lymphatic systems and through defense cells, such as macrophages and leukocytes.

Morphology

The pulpal tissue is traditionally described in histologically distinct, concentric zones: the peripheral odontoblastic layer, the cell-free zone, the cell-rich zone, and the innermost pulp core (Fig 1-29).

The radicular and coronal pulp core is largely *ground substance*, an amorphous hydrated matrix gel that, with collagen fibrils, surrounds and supports the pulpal cells and the vascular and sensory elements. The matrix gel, like interstitial fluid, serves as a transfer medium for transporting nutrients and by-products between widely separated cells of the pulp core and the vasculature. Arterioles and venules, myelinated and unmyelinated nerves, and lymphatic channels, bundled into trunks, pass into and out of the pulp core through the apical foramen or foramina. Collagen is a component of all connective tissues, but in the pulp, the discrete bundles of fibers are dispersed, rather than organized into a supportive framework as in dentin. The proportion of collagen increases with age and is concentrated in the radicular pulp; this facilitates its removal with a barbed broach if a pulpectomy is required.

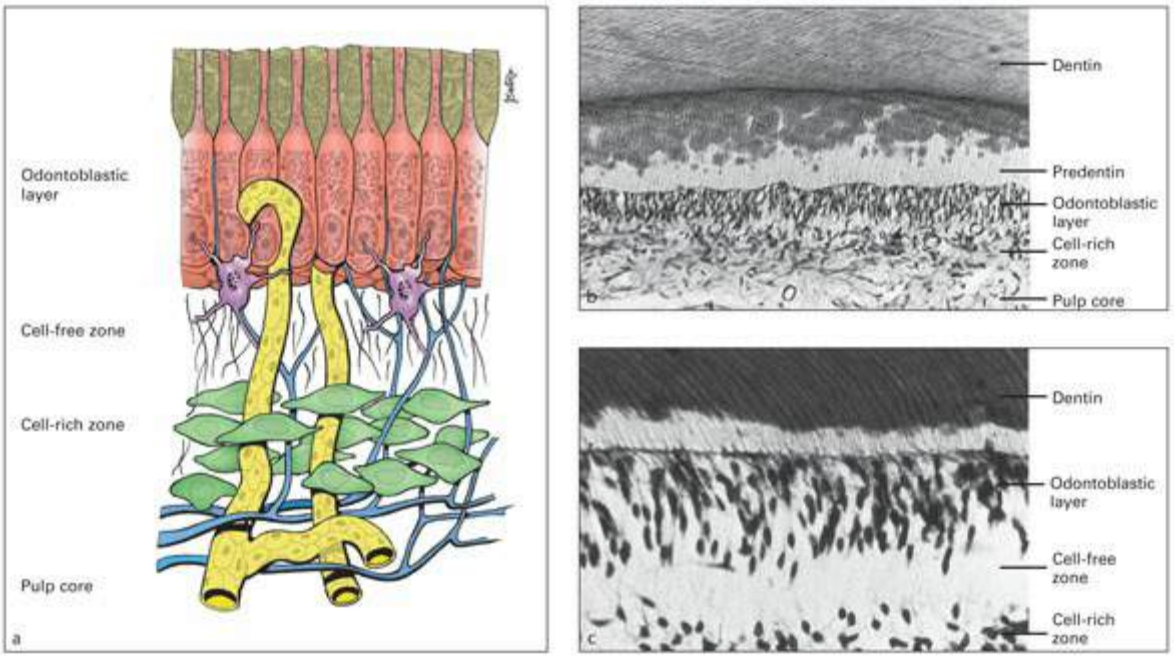


Fig 1-29*(a)* Pulpal histology: odontoblastic layer with immunocompetent dendritic cells, cell-free zone filled with both nerve and capillary plexuses, cell-rich zone of fibroblasts and undifferentiated cells, and pulp core. *(b and c)* Photomicrographs of dentin and underlying pulpal tissues. The multilayered appearance of odontoblast cells is an artifact. (Courtesy of Charles Cox, Birmingham, Alabama.)

The cell-rich zone consists of fibroblasts and undifferentiated cells. The cells of the stratified odontoblastic layer and of the cell-rich zone are separated and supported by a plexus of capillaries and axons that form the cell-free zone. Cells of the pulp function for matrix production, immune reactions, defense, vascular control, and inflammatory response.

Pulpal cells

Odontoblast cell bodies form the outer periphery of the pulp tissue. These specialized, postmitotic cells, each with a process extending into a dentinal tubule, form a single layer at the predentin-pulp interface. As long as the tooth remains vital, they produce and adapt the dentin matrix, including collagen, and they may provide an active transport of calcium ions.[207] The odontoblasts synthesize and secrete various noncollagenous proteins that affect the structure and mineralization of the dentin. Also synthesized are bioactive growth and signaling molecules that supplement neuropeptides within the pulp to coordinate pulpal-dentinal responses for healing and repair. The cellular morphology reflects the stage of activity: large, complex, and columnar when active; small and flattened when quiescent, injured, or aged. Each cell body forms a loose “plug” at the pulpal terminus of the dentinal

tubule and is therefore subject to the hydrostatic shear currents of the tubular fluid. The hydrodynamic effects range from "aspiration" of the cell bodies into the tubules and autolysis of the cells to minor displacement and depolarization of nerve terminals in close contact with the cells. The cell membranes are bound together through a variety of membrane junctions characterized as tight and permanent or gapped and adhesive.[208] Prominent fibers interconnect the cells to form a terminal web, which may orchestrate their secretory functions and stimulus responses as a unified zone. The membrane junctions may modulate either a physiologic barrier or a molecular sieve to regulate transfer of ions and plasma molecules between the interstitial fluid of the pulp and the tubular fluid.[209] Either injury or routine operative procedures can temporarily disrupt the odontoblastic barrier to permit infusion of plasma proteins and to increase outward tubular flow.[210,211]

Fibroblasts, the most numerous pulpal cells, form a dispersed but interconnecting network through their cytoplasmic extensions. They produce, maintain, and remodel pulpal matrix and collagen. They are concentrated in the cell-rich zone supporting the odontoblastic cells. Fibroblasts, undifferentiated mesenchymal cells, stem cells, pericytes, and smooth muscle cells supporting the capillary walls are all prime candidates to be progenitor cells capable of differentiation and phenotype conversion into matrix-secreting replacements for destroyed primary odontoblasts.[212]

Immunocompetent cells include macrophages, lymphocytes, and dendritic cells that function as a host defense system against foreign bodies and antigens.[118,206] *Macrophages* are large cells that are scavengers, able to phagocytize microorganisms, cellular debris, and damaged extracellular matrix. *Dendritic cells* are highly motile cells with branched extensions up to 50 μm long; they have great capacity for encoding antigens on their membrane surfaces. With early caries lesions or injury, they congregate among layers of injured or destroyed odontoblasts and even extend their branches into the affected tubules for surveillance of protein antigens.[213,214] Once antigens are captured, they migrate to nearby lymph nodes to present their encoded antigens to T-lymphocytes. This initial phase is the primary immune response in which memory T cells carrying the antigen "blueprint" are cloned and released back into the pulp tissue. After 3 to 5 days, following another antigen exposure, encoded dendritic cells or macrophages directly interface with the preprogrammed memory T cells in the pulp tissue to release proinflammatory cytokines.[215] With dental caries, this secondary phase of the cell-mediated immune response typically produces chronic inflammation because bacterial by-products diffuse into the pulp long before the organisms themselves directly reach and infect

the tissue.[168] Tubular sclerosis, tertiary dentin, and restorative treatment limit or eliminate the antigen stimulus, so an area in which there is chronic inflammation should revert to a healthy histologic state. However, if microorganisms penetrate the tertiary dentin that forms beneath the caries lesion, the host responds with a prompt and massive influx of neutrophilic and mononuclear leukocytes typical of an acute inflammatory response.[216]

Vascular system

The microvascular system of the pulp[217] contains vessels no larger than arterioles and venules. The primary function is maintenance of tissue homeostasis. The extent of the presence of true lymphatic vessels in normal pulp tissues is somewhat controversial[218,219]; lymphatic vessels exist in most tissues and serve to return tissue fluid and high–molecular weight plasma proteins back to the vascular system. It has been suggested by some investigators that true lymphatic vessels may not be present in all healthy pulpal tissue but may be formed only following inflammation. It has also been hypothesized that in some normal pulpal tissues, the interstitial fluids could simply flow via nonendothelialized interstitial channels and exit the tooth through the apical foramina.[219,220] Capillaries supply oxygen and nutrients that dissolve in and diffuse through the viscous ground substance of the pulp to reach the cells. In turn, the circulation removes waste products, such as carbon dioxide, byproducts of inflammation,[166] and diffusion products that have permeated through the dentin so that they do not accumulate to toxic levels[221] (see Fig 1-14). The equilibrium between diffusion and clearance may be temporarily threatened by use of long-acting anesthetic agents that contain vasoconstrictors, such as epinephrine. An intraligamental injection of a canine tooth with 2% lidocaine with 1:100,000 epinephrine will cause pulpal blood flow to cease for 20 minutes or more.[222] Fortunately, the respiratory requirements of mature pulpal cells are so low that no permanent cellular damage ensues.

Inflammation, the normal tissue response to injury and the first stage of repair, is somewhat modified by the pulp's unique location within the noncompliant walls of the pulp chamber. A stimulus that produces cellular damage initiates neural and chemical signals that increase blood flow and capillary permeability. Plasma proteins, fluids, and leukocytes spill into the confined extracellular space and elevate interstitial fluid pressure.[223] The smooth muscle sphincters that regulate capillary blood flow are under the control of both neurons and local cellular conditions so that a localized vascular response to stimuli may occur independent of

the overall system. Theoretically, elevated extravascular tissue pressure could collapse the thin venule walls and start a destructive cycle of restricted circulation and expanding ischemia. However, the pulpal circulation is unique because it contains numerous arteriole "U-turns," or reverse flow loops, and arteriole-venule anastomoses, or shunts, to bypass the affected capillary bed.[224] Many of these capillaries are normally nonfunctional but facilitate instantaneous localized hyperemia in response to injury. Also, at the periphery of the affected area, where high tissue pressure is attenuated, capillary recapture and lymphatic adsorption of edematous fluids are expedited.[225] These processes confine the edema and elevated tissue pressure to the immediate inflamed area. Animal studies indicate that tissue pressure in an area of pulpal inflammation is two to three times higher than normal but is diminished to nearly normal levels approximately 1.0 mm from the affected area.[226]

Another protective effect of elevated but localized pulpal tissue pressure is a vigorous outward flow of tubular fluid to counteract the pulpal diffusion of noxious solutes through permeable dentin.[227,228] However, an inflammatory condition and higher tissue pressure may also induce *hyperalgesia*, a lowered threshold of sensitivity of pulpal nerves. Thus, an afflicted tooth exposed to the added stress of cavity preparation and restoration may become hypersensitive to cold or other stimuli.[200]

Innervation

Some patients seek dental services because their teeth, through caries lesion(s), injury, or exposed cervical dentin, are painful. Various noxious, thermal, electrical, and mechanical stimuli may be interpreted to some degree as pain. However, pain perception, termed *nociception*, is less important to preservation of pulp vitality than is the role of neuromediation of vascular, inflammatory, immune, and defense functions.[196]

Nociception

Innervation of the pulp[229] is primarily from sensory (afferent) axons, with their cell bodies located a great distance away in the trigeminal ganglion. There are also sympathetic (efferent) axons, with nuclei in the cervical sympathetic ganglia, which produce vasoconstriction when activated. Nerves are classified according to purpose, myelin sheathing, diameter, and conduction velocity. Although a few large

and very high-conduction-velocity A-β nerves with a proprioceptive or touch pressure function have been identified, most sensory interdental nerves are either A-δ nerves or smaller, unmyelinated C fibers. About 13% of the nerves innervating premolars are myelinated A nerves,[230] but they gradually lose their myelin coating as they form the sensory plexus of the cell-free zone and branch into multiple free-end terminals.[231] Up to 40% of the tubules at the pulp horns contain neural filaments extending up to 200 μm into inner dentin,[232] but their role in nociception is unclear. The A-δ nerves have conduction velocities of 13.0 m/sec and low sensitization thresholds to react to hydrodynamic phenomena.[233,234] Activation of the A-δ system results in pain characterized as a sharp, intense jolt.[235]

About 87% of axons innervating premolars are smaller, unmyelinated C fibers, which are more uniformly distributed through the pulp.[236] The conduction velocities of C fibers are slower, 0.5 to 1.0 m/sec, and C fibers are only activated by a level of stimuli capable of creating tissue destruction, such as prolonged high temperatures or pulpitis. The C fibers are also resistant to tissue hypoxia and are not affected by reduction of blood flow or high tissue pressure. Therefore, pain may persist in anesthetized, infected, or even nonvital teeth.[237,238] The sensation resulting from activation of the C fibers is a diffuse burning or throbbing pain, and the patient may have difficulty locating the affected tooth.[235]

Neuromediation of pulpal functions

The close proximity of terminal sensory fibers to odontoblast cells responding to hydrodynamic fluid movement accounts for dentin sensitivity, but the central nervous system is not the sole terminus. As with the odontoblast cells, multiple branches from the axon form the same close apposition with fibroblast cells, perivascular cells, and immunocompetent cells. The terminal nerve ends contain receptors for released or generated bioactive factors caused by injury to the dentin, pulpal cells, or the interstitial environment (see Fig 1-28b). Examples include released dentin growth factors for tertiary dentinogenesis or nerve growth factor (NGF) from fibroblasts to activate sprouting and new growth of additional neurons.[239] Appropriate signals from nearby cells (paracrine), through the blood supply (endocrine), or even from the same cell (autocrine) trigger the release of potent neuropeptides stored within the terminal neurons. Two potent neuropeptides with receptor sites on vascular cells are calcitonin gene–related peptide (CGRP) and substance P; these induce vasodilation and increased blood flow to commence inflammation.[240,241] The importance of the neurogenic link was demonstrated when

teeth that were experimentally denervated and exposed to trauma showed significantly greater pulpal damage because they lacked sufficient inflammatory response.[242] The neuropeptides are so potent that the experimental stimulation of a single C fiber resulted in a detectable increase in blood flow.[243] The exchanges of factors and neuropeptides regulate the inflammatory, dentinogenic, and immunologic functions and other defenses of the dentin-pulp complex. The responses are dynamic, increasing or decreasing with the severity of the external stimulus and with the phase of response (acute, chronic, or healed). These effector functions of the sensory neurons are critical to the function and vitality of the dentin-pulp complex.

Restorative dentistry and pulpal health

Mechanically cutting tooth structure, especially dentin, during restorative treatments generates considerable physical, chemical, and thermal irritation of the pulp. However, if the dentist uses an acceptable and conservative technique and achieves bacterial control, even a mechanical pulp exposure or use of acidic restorative materials poses few problems for pulpal health.[244–247] With all restorative surgical procedures on tooth structure, maintenance of a thick RDT will decrease the chance of pulpal irritation or injury.[248] Although microleakage around restorations is ubiquitous, the fact that almost all pulps remain healthy is related to diminished virulence of the bacteria, relative impermeability of the dentin, and the healing potential of the pulp. An animal study reported 15% pulpal necrosis after a 5.5°C increase in intrapulpal temperature and up to 60% necrosis after an 11°C increase.[249] In vitro crown preparation studies without water coolant record increased intrapulpal temperatures of this degree.[250] Although a retrospective radiographic study of teeth prepared for crowns using only air coolant reported good success,[251] many authorities state that water coolant and intermittent rotary instrument contact with tooth structure during crown preparations is essential to avoid histopathologic damage.[252,253]

Materials utilized in the treatment and restoration of dental tissues can have a direct effect on the pulpal tissues. The main factor influencing the extent of the effect that these materials may exert on the dental pulp is the amount of the RDT. With a reduced RDT, the permeability of the dentin increases,[110] and thus it will more freely allow the penetration of various components of restorative materials into the pulp.[254] In vitro studies have introduced concern that some materials used in adhesive bonding techniques contain various components, such as monomers and acids, that may have deleterious effects on the health and function of pulpal

cells.[255–257] However, animal studies and clinical investigations have shown that many of these materials exhibit acceptable biocompatibility from a clinical aspect when not in direct contact with the pulp tissue.[258–260] Beyond the effects of chemical components of restorative materials, some materials are capable of creating damage to pulpal tissue due to high temperatures from exothermic reactions. Certain methacrylate-based provisional crown materials can generate temperatures during polymerization that are high enough to injure pulp tissue.[261] Tooth-whitening procedures that utilize various oxidizing agents to break down stains in tooth structures have also presented concerns in regard to their effects on the pulpal tissues. Studies have shown that higher concentrations of hydrogen peroxide can diffuse into the pulp and cause damage to the tissues.[262] There also are concerns about the use of certain high-energy light-curing units and the possibility of heat transfer to the pulp during photopolymerization of bonded restorative resins.[263,264]

The aged tooth is less able to respond to noxious stimuli and injury. Age-related changes include reduced blood supply, a smaller pulp chamber, 50% reduction of pulp cells with a lower ratio of pulp cells to collagen fibers, loss and degeneration of myelinated and unmyelinated nerves, decreased neuropeptides, loss of water from the ground substance, and increased intrapulpal mineralizations (denticles).[93,265] However, the aged tooth is generally less sensitive and is protected by sclerotic and tertiary dentin that make it impermeable to diffusion of injurious agents. The dentin-pulp complex is neither static nor noncompliant but is dynamic and adaptive to environmental stresses. Advances in clinical dentistry such as effective preventive measures, sensitive diagnostic tools, improved bonding systems and restorative materials, and conservative surgical preparations should extend the durability and biocompatibility of dental services to preserve the teeth for a lifetime. See chapter 6 for more on pulpal considerations in relation to restorative dentistry.

Potential for regenerative therapies

The regenerative powers of the dental stem cells and undifferentiated mesenchymal cells found in the pulp and other dental tissues have tremendous potential for future dental therapies. Research in the area of dental tissue regeneration involving these progenitor/stem cells is progressing rapidly. These progenitor cells can be found in several different dental tissues, such as the pulp of both exfoliated deciduous and permanent teeth, the periodontal ligament (PDL), the tips of developing roots, and from the tissue (dental follicle) that surrounds the unerupted tooth. All of these progenitor cells probably share a common lineage of being derived from neural crest

cells, and all have generic mesenchymal stem cell–like properties.[266] These mesenchymal stem cells have demonstrated the capability to regenerate dentinal, pulpal, and other tissues that are derived from mesenchymal-type cells. In addition to the mesenchymal progenitor cells that have been identified, a study has recently suggested that epithelial-type stem cells also might exist in the dental pulp of human primary teeth and could play a role in the development of a process for the repair or regeneration of tooth enamel.[267]

Techniques known as *cell homing* or *cell transplantation* are evolving as possible methods to regenerate pulpal tissue within the pulp chamber of a tooth. Some of these methods have shown the ability to regenerate a vascularized pulpal tissue within the confines of a mouse tooth, and the development of odontoblasts that produce a continuous layer of dentinlike tissue lining the dentinal wall of the pulp canal has been demonstrated.[268,269] The ability to regenerate an entire tooth from stem cells has also been investigated. The results of animal studies indicate that the process for regeneration of the various dental tissues required to regenerate a tooth is indeed possible.[270,271] However, there are significant technical difficulties that first must be overcome, such as the development of precise shapes, sizes, and interfaces for the different types of dental tissues.[272] As mentioned earlier in the discussion of the formation of tertiary dentin, research has also indicated the possibility of using a directed repair method that induces pulpal mesenchymal stem cells to repair dentin damaged by disease or trauma. Several experimental techniques have been investigated involving transplantation of various pulp cells, bone marrow cells, or extracellular matrix molecules that can differentiate into odontoblasts and which have the potential to direct a regenerative repair of damaged tissues.[183,273]

Progenitor/stem cells that reside in the PDL have been demonstrated to have the ability to induce alveolar bone formation, remodeling, and the regeneration of PDL tissue.[274,275] It has been shown that these PDL progenitor cells can assemble new PDL-like structures in vivo.[276] Clinical implications of this may include the ability to place implants with PDL progenitor cells applied to the surface that will regenerate both bone and a PDL in the interface between the implant and the bony socket. A clinical study performed on human subjects created what were termed as "ligaplants," which were implants coated with PDL-derived cells. These ligaplants were inserted into the subjects' jaws, and the cells were shown to produce a periodontal attachment in the implant site.[277]

Although research directed toward the clinical regeneration of dental tissues is

still in its early stages, the results of recent studies suggest that the possibility of tissue engineering of human dental tissues in situ is promising. The techniques for the practical applications of these technologies are still being investigated, but the results may have a significant impact on the future of restorative dentistry.

Gingiva

The gingival complex[278,279] forms the interface between the hard, mineralized structure of the tooth, the oral cavity, and the underlying periodontal tissues. As such, its structure and composition are designed to withstand functional forces, such as mastication and toothbrushing; to resist chemical effects of food, drink, and biofilms; and to protect against oral pathogens. Along with tooth-related characteristics, the position, color, contour, and symmetry of the gingiva define esthetic form. High or low gingival margins or red, swollen gingival tissues produce readily visible esthetic compromises. Periodontal disease affects up to 50% of the population in the United States.[280] Gingivitis, the most prevalent form, is characterized as a chronic but reversible inflammatory response of the gingiva to bacterial biofilm.[280] Gingivitis progresses to periodontitis, a more severe and destructive form of periodontal disease, characterized by a loss of PDL and alveolar bone supporting the teeth.[280,281] In general, gingivitis can be reversed and periodontitis can be controlled by the regular removal of supragingival and accessible subgingival biofilm.[282,283] When restorations are required, a healthy gingival response can be ensured by the clinician's insertion of restorations that precisely replicate the contours and surface smoothness of the tooth structure they are replacing, along with meticulous oral hygiene on the part of the patient.[284,285]

Morphology

A normal, healthy gingiva (Fig 1-30) presents a scalloped marginal outline, firm texture, coral pink or normally pigmented coloration (depending on ethnicity), and, in about 40% of the population, a stippled surface. Coronally, free gingiva includes the scalloped cuff of tissue forming the marginal crest, which curves internally to form a narrow internal crevice or sulcus around the tooth. The external free gingiva extends apically 1.0 to 2.0 mm to the free gingival groove, corresponding to the base of the sulcus and located at the level of the CEJ in a healthy situation. The main component of the gingiva, the *attached gingiva*, is firmly affixed to the periosteum of the alveolar bone and hard palate and to the supra-alveolar cementum of the root of

each tooth. The free and attached gingiva extends interproximally to form the *interdental papilla* that fills the gingival embrasure below the interproximal contacts. Figure 1-31 shows the various collagen fiber bundles that attach the tooth and gingival tissue to the bone and to each other. In a healthy periodontium in which the gingival tissue and bone have not receded apically, the interdental papilla completely fills the gingival embrasure. However, if the periodontal health is neglected, the interdental papilla may not completely fill the gingival embrasure. Unlike attached gingiva, the epithelial lining of the sulcus and of the papillary col between the facial and lingual sides of the papilla is not keratinized. Therefore, in a mouth that harbors periodontal pathogens, without effective oral hygiene procedures, both the sulcus and the papilla are susceptible to inflammatory reactions to the plaque biomass that accumulates there. The *vertical width* or *zone* of keratinized gingiva refers to the distance from the free gingival margin to the mucogingival junction. The *mucogingival junction* is the junction of the keratinized gingiva with the alveolar mucosa, which is mobile, darker red, and nonkeratinized. The usual width of keratinized gingiva varies by location on both the facial and lingual aspects of the teeth. Facially, it is normally widest in the incisor areas and narrowest in the canine and first premolar regions. Lingually, it varies from less than 2.0 mm in the area of the mandibular incisors to 9.0 mm on the lingual aspects of mandibular molars.[287]

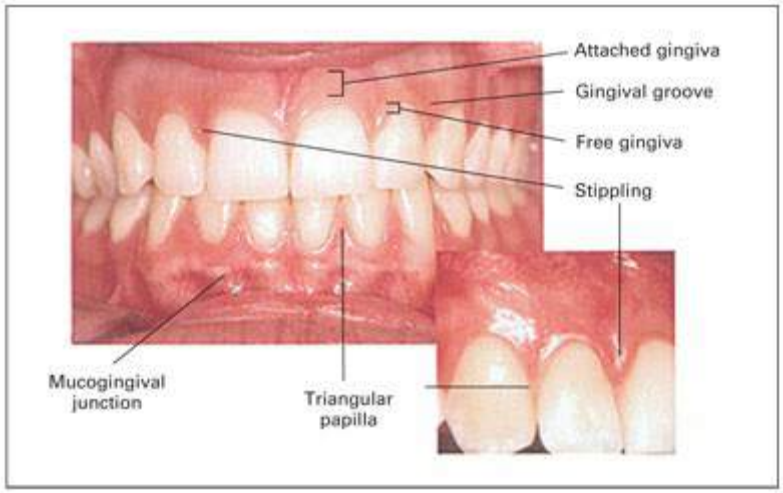


Fig 1-30 Clinically healthy, normal gingiva. (Reprinted from Chapple and Gilbert[286] with permission.)

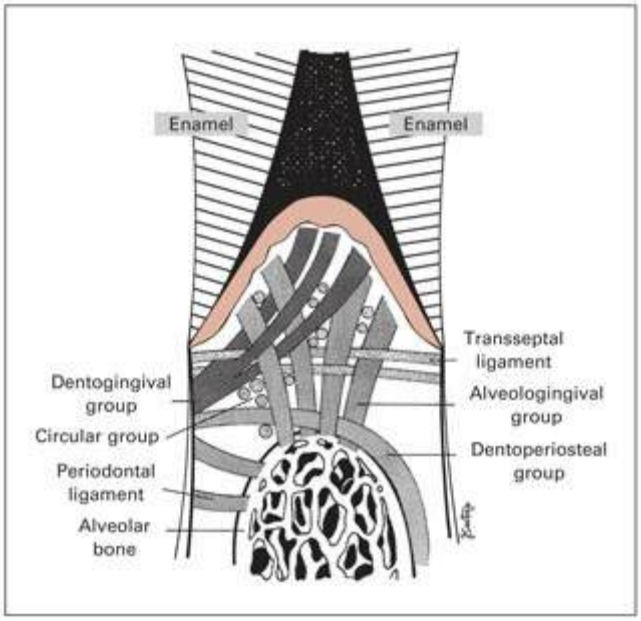


Fig 1-31 Interproximal papilla and attached gingiva. Various collagen fiber bundles continuous with the connective tissue attachment circle the teeth or attach the gingiva to the cementum and bone. The gingival embrasure or interdental space below the contact is prone to plaque accumulation, which is responsible for both caries lesion development and gingivitis.

The significance of the width (zone) of keratinized gingiva in restorative dentistry is somewhat controversial. Lang and Löe[287] evaluated sites without restorations in adults with effective oral hygiene and concluded that a minimum width of 2.0 mm of keratinized gingiva is required to prevent chronic gingival inflammation. Maynard and Wilson[288] recommended 5.0 mm of keratinized gingiva (2.0 mm of free and 3.0 mm of attached gingiva) to achieve predictability of gingival response to restorations with margins placed within the gingival crevice or sulcus. These same authors also advised that the thickness of the gingiva be evaluated. In clinical situations in which the tissue is thin enough to see the periodontal probe through the free gingival margin, the soft tissues may be unable to support intracrevicular restorative procedures.[288] In a human clinical study, Stetler and Bissada[289] compared areas of a narrow (less than 2 mm) zone of keratinized gingiva to wide (greater than 2 mm) zones. In sites with no restorations present, there was no clinical difference in gingival inflammation between narrow and wide areas. Teeth in narrow keratinized areas with subgingival restorations had greater inflammation.[289] In a similar study,[290] restored teeth with margins placed within the gingival sulcus adjacent to narrow areas of keratinized gingiva suffered increased recession and attachment loss. In summary, both the width (zone) and thickness of the gingival tissues should be evaluated prior to the placement of restorations that will extend subgingivally. In compromised areas, mucogingival therapy (soft tissue grafting)

prior to placing restorations should be considered.

Dentogingival complex and biologic width

The junctional epithelium and the subjacent connective tissue between the base of the gingival sulcus and the alveolar crest stabilize and seal the gingiva around the cervical enamel and supra-alveolar cementum surfaces. The combination of these two tissues (junctional epithelium and connective tissue attachment) is termed the *dentogingival junction*, and their combined vertical dimension is termed the *biologic width* (Fig 1-32). The junctional epithelium (or epithelial attachment) is nonkeratinized and provides intimate adhesion against the cervical enamel or, with gingival recession or attachment loss, against the cementum. With inflammation, a periodontal probe used to measure pocket depth may easily penetrate the junctional epithelium to the level of the connective tissue attachment. The *clinical attachment level* is the distance from the CEJ to the tip of the probe when the probe is in the sulcus. As a defensive mechanism against pathogenic bacterial penetration, the junctional epithelial cells have a high mitotic rate and are rapidly exfoliated and replaced. The cellular spacing and narrow, tapering width of the junctional epithelium layer facilitates the movement of *crevicular fluid*, a serum exudate containing defensive cells, complement, and antibodies, into the sulcus. These same characteristics of the epithelium also allow bacterial antigens and toxins to pass from the sulcus into the underlying tissues. The inflammatory response may become exaggerated and result in gingival edema, color changes, and bleeding, classic signs of gingivitis. The amount of crevicular fluid exudate and of bleeding with probing are two indices of gingival inflammation.[286,291]

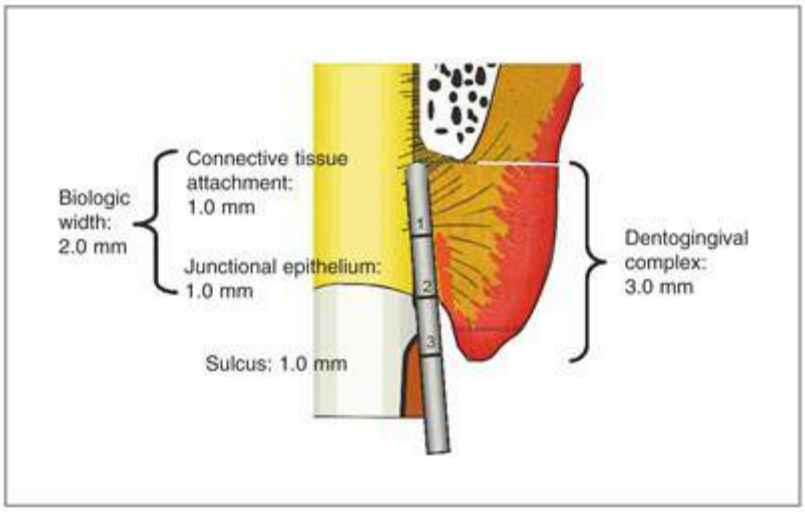


Fig 1-32 Biologic width and dentinogingival complex. Note that the gingival crown cavosurface margin is ideally no more than 0.5 mm into the sulcus (no closer than 2.5 mm from the osseous

crest). The tip of the periodontal probe has been pushed through the DEJ (junctional epithelium and connective tissue attachment) to the osseous crest (bone sounding).

The protective body defenses can be overcome by the chronic accumulation of pathogenic bacteria in the biofilm or at local plaque-retaining sites, such as calculus or imperfect restoration margins. With unfavorable genetic host factors, the inflammatory response can become destructive. Proliferation and apical migration of the junctional epithelial cells create greater pocket depths and a loss of attachment with resorption of alveolar bone, both manifestations of periodontitis. The internal and external epithelial layers are supported by a network of collagen fiber bands coursing around and between the teeth and anchoring the gingiva to the alveolar bone and cementum of the roots (see Fig 1-31). From periodontally healthy adult cadaver dissections, Vacek et al[292] reported that the mean vertical dimension of the junctional epithelium was 1.14 mm. The subjacent connective tissue attachment measured a relatively consistent 0.77 mm. Together, the mean biologic width dimension was 1.91 mm (1.75 mm for anterior teeth and 2.08 mm for posterior teeth). The results affirm an earlier dissection study by Gargiulo et al[293] in which the biologic width averaged 2.04 mm. Based on these dissections and on clinical experience, clinicians have generally accepted a 1.0-mm depth guideline for a healthy gingival sulcus. Thus, at a midfacial location, 3.0 mm (2.0 mm biologic width plus an additional 1.0-mm sulcus depth) is a simplified working concept of the average dimensions of the dentogingival complex (see Fig 1-32). Interproximally, with adjacent teeth present, the dentogingival complex dimension is reported to be in a range of 3.0 to 4.5 mm.[294,295]

Several clinical studies have demonstrated that the closer a crown margin is to the dentogingival attachment (ie, the deeper into the gingival sulcus the margin is placed), the greater the probability of an inflammatory response, as evidenced by increased gingival plaque, increased bleeding indices, and, with time, the loss of attachment.[296–298] A prospective clinical study[299] evaluating 480 metal-ceramic crowns found that the risk of gingival bleeding was related to the baseline oral hygiene index and was twice as high for intrasulcular margins as for supragingival margins. Thus, it is generally accepted that a supragingival crown margin is beneficial for periodontal health. However, a subgingival preparation margin is often necessary because of a short clinical crown, cervical or root caries lesions, a coronal fracture, or the need to hide the margin. Maintenance of the 2.0-mm dimension of the dentogingival junction is generally considered necessary for maintaining periodontal health. Although patient susceptibility varies, the assumption

is that a violation of the biologic width, especially into the connective tissue attachment, will lead to chronic inflammation and perhaps the loss of attachment, bone resorption, and gingival recession.[300]

It is difficult to clinically measure the separate tissue components that make up the biologic width. Kois[294,295] advocates the diagnostic use of a periodontal probe to confirm the level of the bone support to validate a biologically safe placement of the gingival margin. With anesthesia, the probe penetrates the midfacial sulcus to contact the osseous crest, a process called *bone sounding* (see Fig 1-32). The distance from the free gingival margin to the osseous crest defines the *dentogingival complex* (ie, the biologic width plus the sulcus depth). In 85% of subjects, a 3.0-mm reading confirms existence of an optimum biologic width, and the preparation margin may be safely placed 0.5 mm below the existing gingival margin. In addition, the preparation must follow the contours of the gingival margin, which parallels the CEJ, to avoid violation of the biologic width (or biologic zone), especially in the interproximal areas. With maxillary incisors, the proximal CEJ and junctional attachment may be scalloped as much as 3.5 mm incisal to the level of the facial CEJ (Fig 1-33). A midfacial reading greater than 3.0 mm confirms distant osseous support for the gingiva so that recession of the gingiva or interdental papillae may result following restorative procedures. An important anatomical study has demonstrated a quantitative relationship between the height of the gingival papilla and the distance from the alveolar crest to the base of the proximal contact. At a distance of 5.0 mm, 98% of the dental papillae completely filled the restored gingival embrasure; but at 6.0 mm, only 56% did so.[301] The dentist might be able to adjust the incisal/occlusal–gingival level of restored proximal contacts accordingly, in order to avoid an unesthetic space between the tip of the papilla and the proximal contact. If the dimension of the dentogingival complex is less than 3.0 mm, which suggests a compromised biologic width, a mucogingival flap and osseous resection (a crown lengthening procedure) is generally recommended to achieve the necessary dentogingival complex dimension in the area.[302]

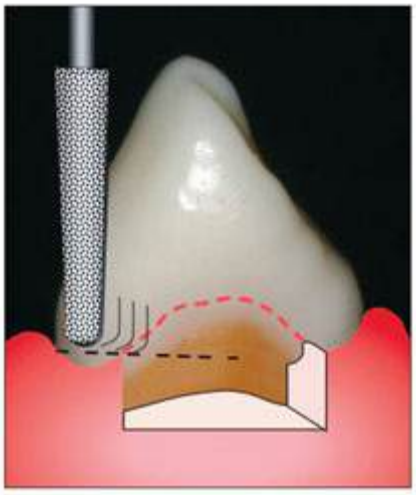


Fig 1-33 The dentist should carefully parallel and duplicate the pronounced curve of the interproximal gingiva (and CEJ) during crown preparation, as shown by the *pink dotted line*, to prevent violation of the biologic width.

The clinical imperative of a 2.0-mm biologic width is based on the averages of dimensions determined in the autopsy studies cited above; however, much variation was found in the dimensions of the individual tissues, particularly the depth of the sulcus and the length of the junctional epithelium.[292,293] Fifteen percent of teeth in the study by Vacek et al[292] had restoration margins less than 2.0 mm from the osseous crest with no related loss of attachment. In an animal study, Class 5 restorations were placed directly at the osseous crest. At 1 year, a functional biologic width re-formed apically with one-fifth the linear dimensions of the preoperative nonrestored controls.[303] A 2-year prospective clinical study evaluated the periodontal effects of subgingival crown margins within the dentogingival complex in three groups of patients: Group 1 consisted of patients with subgingival crown margins 1.0 mm or less from the alveolar crest (within the connective tissue attachment); Group 2 included patients with margins between 1.0 and 2.0 mm from the alveolar crest (within the junctional epithelium); and Group 3 included patients with margins 2.0 mm or greater from the alveolar crest (within the sulcus). A marked increase in papillary bleeding for Group 1 was the most significant sign that the health of the periodontium was compromised. Even though gingival inflammation resulted from the violation of the connective tissue attachment, there were no signs of either gingival recession or bone resorption.[304] On the other hand, a review[305] of the literature on the placement of the gingival margin based on traditional concepts of biologic width concluded that the evidence is primarily based on opinion and anecdotal cases. Nevertheless, the authors suggest that "clinical experience and prudence" favor a 3.0-mm space between the alveolar crest and the restoration

margin.[305] For additional discussion of evaluation and management of tissue during crown preparation, see chapter 18.

Defective restorations and periodontal health

Dental restorations have a definite impact on the health of the periodontal tissues. In a longitudinal study involving 884 individuals, the probing depths and gingival recession were recorded for each individual at age 26 years and again at age 32 years. The results of the study showed that for a periodontal site in which a restoration was placed prior to the age of 26 years, the attachment loss in the corresponding periodontal site at age 32 years would be twice as likely to be ≥ 3.0 mm than if the adjacent tooth surface had remained sound until age 32 years.[306] In addition to the subgingival placement of the restoration margin described earlier, poor quality of the restoration margin, including marginal openings, roughness, and overhangs, may impair periodontal health. With an ideal preparation, it is technically possible to obtain a marginal discrepancy of less than 10 μm with a cast metal restoration and less than 50 μm with a ceramic restoration.[307] However, with additive effects of luting cement thickness, preparation design deficiencies, and technique errors, the interface discrepancy can be much greater. In a retrospective study of 42 crowns with intrasulcular margins and more than 4 years of service, the mean marginal discrepancy was 160 μm, and 15 crowns (36%) had discrepancies greater than 200 μm. The investigators reported a direct quantitative correlation between increased marginal discrepancy and an increased gingival index and crevicular fluid-flow volume.[308] Another study reported that, in addition to increased inflammation, radiographically determined bone loss was associated with crown margin discrepancies greater than 50 μm.[309] With a marginal discrepancy of 200 μm completely surrounding a restored tooth, the total computed area of exposed cement surface amounts to several square millimeters. This interface is rough, porous, and retentive to biofilm.[310]

A common problem with restoration margins, especially with Class 2 direct restorations, is the *overhang*, an extension of the restorative material beyond the cavity preparation. The morphologic variation in the cervical aspect of teeth, including furcations, fluting, convexities, and concavities, makes it difficult to consistently place a wedge and matrix band to fully adapt to the gingival cavosurface margin. Lervik et al[311] magnified bitewing radiographs and found that 25% of proximal-surface restorations presented overhangs. Of these, 29% were greater than 0.2 mm and 4% were greater than 0.5 mm. Jeffcoat and Howell[312] grouped 100

restorations with overhangs by the percentage of intrusion or invasion into the interproximal space and compared adjacent bone loss with the 100 contralateral nonrestored controls. They concluded that gingival displacement caused by large overhangs intruding 51% or more into the interproximal space is directly related to alveolar bone loss. In a literature review of overhanging dental restorations (ODRs) and their effects on the periodontium, Brunsvold and Lane[313] reported a prevalence range of interproximal overhangs from 25% to 76%. The authors attributed the wide range to differences in classification criteria and diagnostic methods. Their conclusions, which follow, are validated by recent studies and reviews[314,315]:

- Prevalence of ODRs is high, involving at least 25% of restored teeth.
- Radiographic and tactile exploration must be combined to improve ODR detection.
- Increased bone loss, attachment loss, pocket depth, and inflammation occur adjacent to teeth with ODRs; the periodontal destruction is related to the size of the ODR.

Restoration overhangs have been described as "permanent calculus." As with poor marginal fit of crowns, iatrogenic defects in restorations contribute to the retention and concentration of plaque biomass. Lang et al[316] found that an ODRs also alter the pathologic nature of the biofilm bacteria. Two sets of MOD onlays were made for each subject, one with normal contours and one with a 0.5- to 1.0-mm overhang. One of the restorations was temporarily cemented, and after 19 to 27 weeks, that restoration was removed and the other was placed in a crossover study design. With the overhanging restorations, a normally healthy sulcus microflora changed to a gram-negative, anaerobic culture typical of chronic periodontitis. It is important to note that when the onlays with overhangs were exchanged for the properly fitted restorations, equivalent to the clinical removal of an overhang, the gingival index and microflora returned to a healthy state. However, in a study in which amalgam overhangs were removed with an ultrasonic scaler and followed for 3 months, baseline gingival inflammation and early bone loss were not significantly reversed except in the group with meticulous plaque control.[317] In another similar study, a combination of the removal of the overhangs, periodontal scaling, and antibiotics resulted in beneficial effects on the probing depths and clinical attachment levels, along with a reduction of the population of some pathologic bacteria.[318] Thus, in addition to encouraging patients' oral hygiene, the clinician should carefully plan the location of restoration margins and strive for excellent fit

and contour of the restoration. When overhangs are detected, their removal should be part of the treatment plan.[319]

Other parameters of restorations that have a less direct detrimental effect on the health of the periodontium are overcontoured axial surfaces, traumatic occlusion, and defective interproximal contacts. Undercontoured or "flat" facial and lingual profiles have been advocated to benefit periodontal health, while overcontoured restorations have been associated with biofilm retention.[320] Crowns are commonly overcontoured, from 0.7 to 1.28 mm wider than the buccolingual dimension of the teeth before restoration.[321] It appears that some latitude in facial and lingual contours is acceptable, but it is important to incorporate "fluting" in areas where the restoration abuts furcations to facilitate plaque removal.[322] A common clinical sign of traumatic occlusion on a specific tooth or group of teeth, especially with a loss of periodontal support, is hypermobility. A clinical study of patients with periodontitis found a relationship between tooth mobility and increased loss of both attachment and bone.[323] In their review, the authors concluded that in patients with preexisting plaque-induced periodontitis, concomitant occlusal trauma may increase the rate of destruction, but "trauma from occlusion cannot induce periodontal tissue breakdown."[324]

Another aspect of restorations that relates to the health and comfort of the periodontium is deficient or open interproximal contacts. Acceptable interproximal contacts should contribute to tooth stability, deflect food to the vestibular or oral embrasures to protect the nonkeratinized gingival col, and prevent food impaction. However, when the level of oral hygiene is good and there is no food impaction, there is no clear relationship between deficient interproximal contacts and periodontal problems.[325] Indeed, the majority of healthy teeth, unless in heavy occlusion, have a slight interproximal gap that will often allow an 8-µm-thick strip of shimstock to pass through the contact.[326] Nevertheless, if the teeth are in contact preoperatively, authorities agree that the new restoration should restore the contact to prevent possible gingival irritation and patient discomfort from food impaction.

Conclusion

Just like the dentition, the periodontium is at risk from specific biofilm-induced destruction; therefore, long-term oral health requires routine effective plaque removal. Errors in diagnosis and treatment planning and iatrogenic problems associated with the restoration of teeth can exacerbate biofilm retention and contribute to inflammation, loss of connective tissue attachment, and loss of bone

support. A healthy periodontium is essential for the well-being, comfort, and esthetics of the oral structures.

Acknowledgment

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